(A)

The vapour pressure N- Heptane and N-Octane (B) are given in the follow table Assume that Raulot's and dalton law and apply. compare vapour liq. equi

Boiling point of N- Heptane is 371.4 K. Boiling point of N- Octane is 398.6 K

T° (C)	98.4	105	110	115	120	125
T° (K)	371.4	378	383	388	393	398.6
P_A° (kpa)	101.325	125.323	139.988	159.987	179.985	205.316
P_B° (kpa)	44.396	55.595	64.528	74.795	88.659	101.325

$P_t = 101.325$ kpa. Construct t-x-y graph.

$\alpha = \frac{P_A^\circ}{P_B^\circ}$

Soln:

$y = \frac{P_A^\circ x}{P_t}$

$$x = \frac{P_t - P_B^\circ}{P_A^\circ - P_B^\circ}$$

$$x_1 = \frac{[101.325] - [44.396]}{[101.325] - [44.396]} = 1$$

$$x_2 = \frac{[101.325] - [55.595]}{[125.323] - [55.595]} = \frac{45.73}{69.728}$$

$$= 0.655$$

$$x_3 = \frac{[101.325] - [64.528]}{[139.988] - [64.528]} = \frac{36.797}{75.46}$$

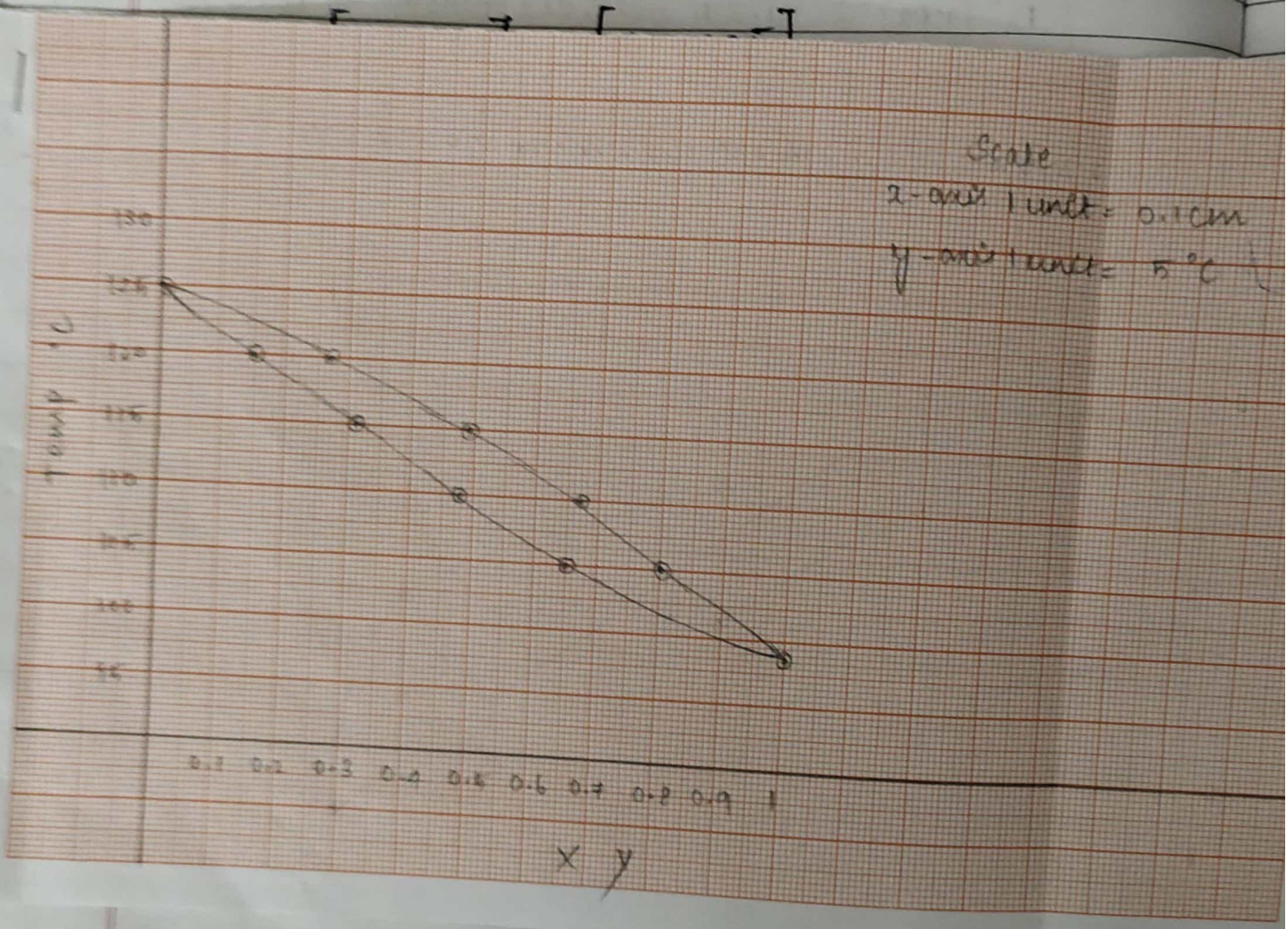
$$= 0.487$$

$$x_4 = \frac{[101.325] - [74.795]}{[159.987] - [74.795]} = \frac{26.53}{85.192}$$

$$= 0.3114$$

$$x_5 = \frac{[101.325] - [86.659]}{[179.985] - [86.659]} = \frac{14.666}{93.326}$$

$$= 0.157.$$



$$\alpha = \frac{P_{A_4}^{\circ}}{P_{B_4}^{\circ}} = \frac{159.987}{74.795} = 2.13$$

$$\alpha = \frac{P_{A_5}^{\circ}}{P_{B_5}^{\circ}} = \frac{179.985}{86.659} = 2.07$$

$$x_4 = \frac{[101.325] - [74.795]}{[159.987] - [74.795]} = \frac{26.53}{85.192}$$

$$= 0.3114$$

$$x_5 = \frac{[101.325] - [86.659]}{[179.985] - [86.659]} = \frac{14.666}{93.326}$$

$$= 0.157$$

$$x_6 = \frac{[101.325] - [101.325]}{[205.316] - [101.325]} = 0$$

$$\alpha = \frac{P_A^\circ}{P_B^\circ}$$

$$\alpha = \frac{P_{A_1}^\circ}{P_{B_1}^\circ} = \frac{101.325}{44.395} = 2.282$$

$$\alpha = \frac{P_{A_2}^\circ}{P_{B_2}^\circ} = \frac{125.323}{55.595} = 2.254$$

$$\alpha = \frac{P_{A_3}^\circ}{P_{B_3}^\circ} = \frac{139.988}{64.528} = 2.169$$

$$\alpha = \frac{P_{A_4}^\circ}{P_{B_4}^\circ} = \frac{159.987}{74.795} = 2.13$$

$$\alpha = \frac{P_{A_5}^\circ}{P_{B_5}^\circ} = \frac{179.985}{86.659} = 2.07$$

$$\alpha = \frac{P_{Ab}^{\circ}}{P_{Bb}^{\circ}} = \frac{205.316}{101.325} = 2.02$$

$$y = \frac{P_A^{\circ} x}{P_t}$$

$$y_1 = \frac{101.325\ (1)}{101.325} = 1$$

$$y_2 = \frac{(125.33)(0.65)}{101.325} = 0.80$$

$$y_3 = \frac{(139.488)(0.487)}{101.325} = 0.672$$

$$y_4 = \frac{(159.987)(0.3114)}{101.325} = 0.491$$

$$y_5 = \frac{(179.985)(0.157)}{101.325} = 0.278$$

$$y_6 = \frac{(205.316)(0)}{101.325} = 0.$$

2. A vapour pressure of n- heptane and n- octane which is given in the table [1] at 101.325 kp assume that Rault's and Dalton's law applied the value of relative volatility generate x, y data and considered x, y plot using the data from 1st question.

α & α_{avg} ⟶ problem ①

$\alpha_{avg} = 2.157$

$x = 0$ to 1

$$y = \frac{\alpha_{avg}\, x}{(1 + (\alpha - 1)\, x)}$$

eq.

$$y_1 = \frac{(2.157)(0)}{1 + (2.157 - 1)\,0} = 0$$

$$y_2 = \frac{(2.157)(0.1)}{1 + (2.157 - 1)\,0.1} = 0.1933$$

$$y_3 = \frac{(2.157)(0.2)}{1 + (2.157 - 1)(0.2)} = \frac{0.4314}{1.2314} = 0.350$$

$$y_4 = \frac{(2.157)(0.3)}{1 + (2.157 - 1)(0.3)} = \frac{0.6471}{1.3471} = 0.480$$

$$y_5 = \frac{(2.157)(0.4)}{1 + (2.157 - 1)(0.4)} = \frac{0.8628}{1.4628} = 0.589$$

0.2157 1.1157

$$y_6 = \frac{(2.157)(0.5)}{1+(2.157-1)(0.5)} = \frac{1.0785}{1.5785} = 0.6832$$

$$y_7 = \frac{(2.157)(0.6)}{1+(2.157-1)(0.6)} = 0.763$$

$$y_8 = \frac{(2.157)(0.7)}{1+(2.157-1)(0.7)} = 0.834$$

$$y_9 = \frac{(2.157)(0.8)}{1+(2.157-1)(0.8)} = 0.896$$

$$y_{10} = \frac{(2.157)(0.9)}{1+(2.157-1)(0.9)} = 0.959$$

$$y_{11} = \frac{(2.157)(1)}{(1+(2.157-1)(1)} = 1$$

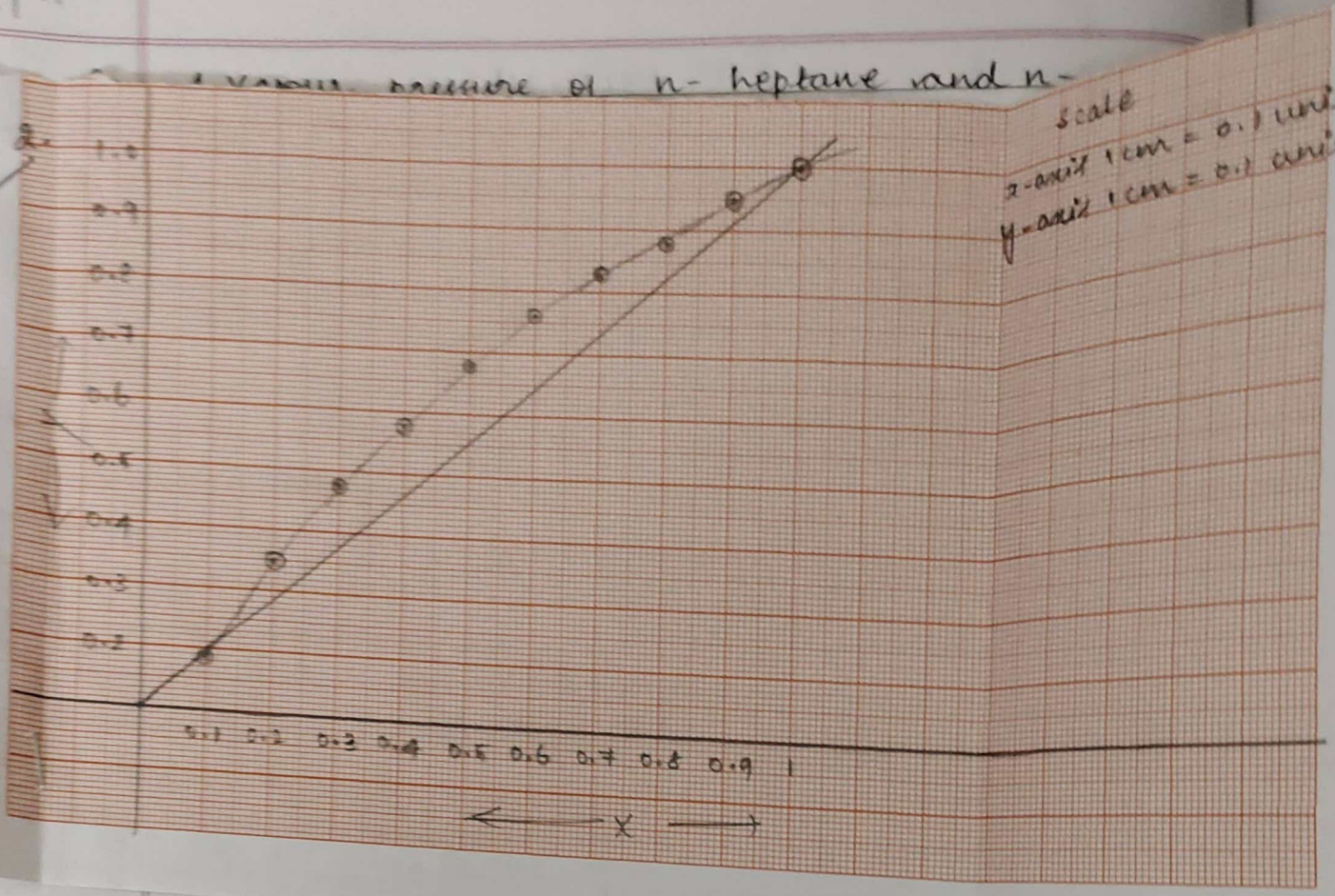
3. Compute the equ. Data @ 760 mmHg pressure. also calculate relative volatality

P_A (mmHg)	760	830	920	1060	1200	1360
P_B (mmHg)	200	350	420	550	690	760

To find x, y, α. αy

4. A solution ethanol & Methanol are substantialy ideal. Compute the vapour liq equi for the sys at 1 atm. pressure & relative volatality & also we assume that Roulat's & Dalton's law apply the value of avg relative volatality. Generate x, y data & construct txy & xy plot.

7/9/22



$$y_2 = \frac{(2.157)(0.1)}{1+(2.157-1)0.1} = 0.1933$$

$$y_3 = \frac{(2.157)(0.2)}{1+(2.157-1)(0.2)} = \frac{0.4314}{1.2314} = 0.350$$

$$y_4 = \frac{(2.157)(0.3)}{1+(2.157-1)(0.3)} = \frac{0.6471}{1.3471} = 0.4803$$

$$y_5 = \frac{(2.157)(0.4)}{1+(2.157-1)(0.4)} = \frac{0.8628}{1.4628} = 0.5898$$

t(°c)	66.53	70	72	74	76	78
P_A (methanol) (mmHg)	760	867.5	934.94	1006.79	1082.79	1163.5
P_B (ethanol) (mmHg)	467.8	541.77	588.66	638.9	692.66	750.14

To find x_A, y_A, x , Find x, y data

5. Compute the vapour liq equil Data from the following vapour pressure data @ 760 mmHg & construct txy plot.

T(°C)	98.4	105	110	120	125.6
P_A (mmHg)	760	940	1050	1350	1540
P_B (mmHg)	333	417	484	650	760

Ans:

Q. Soln:

$$x = \frac{P_t - P_B^\circ}{P_A^\circ - P_B^\circ}$$

$$x_1 = \frac{760 - 200}{760 - 200} = 1$$

$$x_2 = \frac{760 - 350}{830 - 350} = 0.854$$

$$x_3 = \frac{760 - 420}{820 - 420} = 0.68$$

78.53

$$x_4 = \frac{760 - 550}{1060 - 550} = 0.412$$

1177.4

$$x_5 = \frac{760 - 690}{1200 - 690} = 0.137$$

760

$$x_6 = \frac{760 - 760}{1360 - 760} = 0$$

$$y = \frac{P_A^\circ (x)}{P_t}$$

$$y_1 = \frac{760\,(1)}{760} = 1$$

830

$$y_2 = \frac{760\,(0.854)}{760} = 0.933$$

920

$$y_3 = \frac{760\,(0.68)}{760} = 0.823$$

1060

$$y_4 = \frac{760\,(0.412)}{760} = 0.575$$

1200

$$y_5 = \frac{760\,(0.137)}{760} = 0.216$$

1360

$$y_6 = \frac{760\,(0)}{760} = 0$$

$$\alpha = \frac{P_A^\circ}{P_B^\circ}$$

$$\alpha_1 = \frac{760}{200} = 3.8$$

$$\alpha_2 = \frac{830}{350} = 2.37$$

$$\alpha_3 = \frac{920}{420} = 2.19$$

$$\alpha_4 = \frac{1060}{550} = 1.93$$

$$\alpha_5 = \frac{1200}{690} = 1.73$$

$$\alpha_6 = \frac{1360}{760} = 1.78$$

$$\alpha_{avg} = \frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6}{6}$$

$$= 2.303$$

4.

$$x = \frac{P_t - P_B^\circ}{P_A^\circ - P_B^\circ}$$

1 atm = 760 mmHg

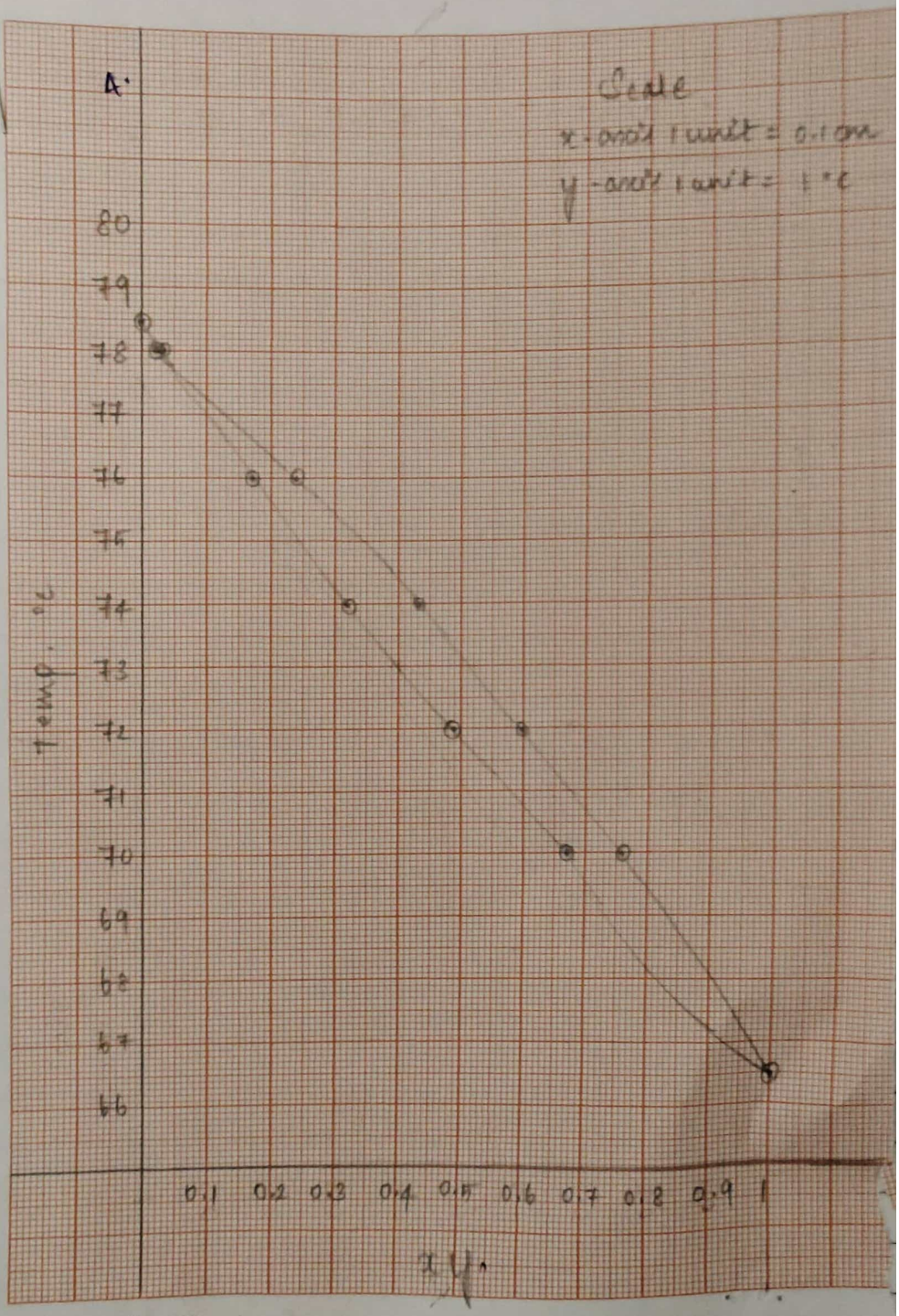
$$x_1 = \frac{760 - 467.8}{760 - 467.8} = 1$$

$$x_2 = \frac{760 - 541.77}{867.5 - 541.77} = 0.67$$

$$x_3 = \frac{760 - 588.66}{934.94 - 588.66} = 0.495$$

$$x_4 = \frac{760 - 638.9}{1006.6 - 638.9} = 0.329$$

$$\alpha_2 = \frac{830}{350} = 2.37$$



$$x_4 = \frac{760 - 638.9}{1006.6 - 638.9} = 0.329$$

$$x_5 = \frac{760 - 692.66}{1082.79 - 692.66} = 0.173$$

$$x_6 = \frac{760 - 750.14}{1163.6 - 750.14} = 0.024$$

$$x_7 = \frac{760 - 760}{1177.4 - 760} = 0$$

$$y = \frac{P_A^\circ (x)}{Pt}$$

$$y_1 = \frac{760\,(1)}{760} = 1$$

$$y_2 = \frac{867.5\,(0.67)}{760} = 0.765$$

$$y_3 = \frac{934.94\,(0.495)}{760} = 0.609$$

$$y_4 = \frac{1006.79\,(0.329)}{760} = 0.436$$

$$y_5 = \frac{1082.79\,(0.173)}{760} = 0.246$$

$$y_6 = \frac{1163.6\,(0.024)}{760} = 0.0365$$

$$y_7 = \frac{1177.4\,(0)}{760} = 0$$

$$\alpha = \frac{P_A^\circ}{P_B^\circ} \qquad \alpha_1 = \frac{760}{467.8} = 1.625$$

$$\alpha_2 = \frac{867.5}{541.77} = 1.601$$

$$\alpha_3 = \frac{934.94}{588.66} = 1.588$$

$$\alpha_4 = \frac{1006.6}{638.9} = 1.576$$

$$\alpha_5 = \frac{1082.79}{692.66} = 1.563$$

$$\alpha_6 = \frac{1163.6}{750.14} = 1.551$$

$$\alpha_7 = \frac{1177.4}{766} = 1.549$$

$$\alpha_{avg} = \frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6 + \alpha_7}{7}$$

$$\alpha_{avg} = 1.579$$

$x = 0$ to 1

$$y = \frac{\alpha_{avg}\, x}{(1 + (\alpha - 1)x)}$$

$$y_1 = \frac{1.579\,(0)}{1 + (1.579 - 1)\,0} = 0$$

$$\frac{0.1579}{1.0579} = 0.149$$

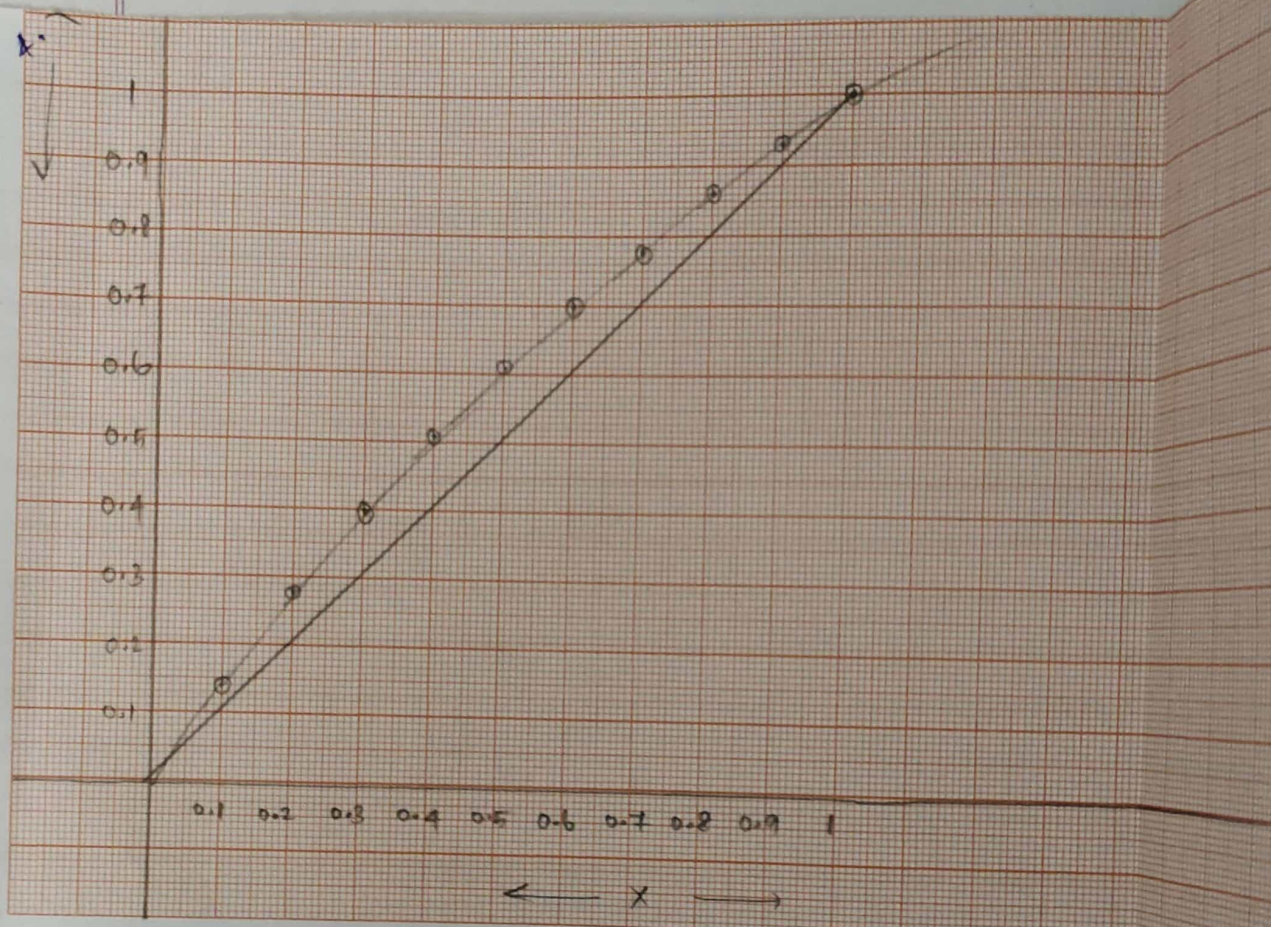
$$y_2 = \frac{1.579\,(0.1)}{1 + (1.579 - 1)(0.1)} = \frac{0.1579}{1.0579}$$

$$y_3 = \frac{1.579\,(0.2)}{1 + (1.579 - 1)(0.2)} = \frac{0.3158}{1.1158} = 0.283$$

$$\alpha_3 = \frac{934.94}{588.66} = 1.588$$

$$\alpha_4 = \frac{1006.6}{638.9} = 1.576$$

$$\alpha_5 = \frac{1082.79}{692.66} = 1.563$$



$$y_2 = \frac{1.579\ (0.1)}{1+(1.579-1)(0.1)} = \frac{0.1579}{1.0579}$$

$$y_3 = \frac{1.579(0.2)}{1+(1.579-1)(0.2)} = \frac{0.3158}{1.1158} = 0.283$$

$$y_4 = \frac{1.579(0.3)}{1+(1.579-1)(0.3)} = \frac{0.4737}{1.1737} = 0.403$$

$$y_5 = \frac{1.579(0.4)}{1+(1.579-1)(0.4)} = \frac{0.6316}{1.2316} = 0.512$$

$$y_6 = \frac{1.579(0.5)}{1+(1.579-1)(0.5)} = \frac{0.7895}{1.2895} = 0.6122$$

$$y_7 = \frac{1.579(0.6)}{1+(1.579-1)(0.6)} = \frac{0.9474}{1.3474} = 0.703$$

$$y_8 = \frac{1.579(0.7)}{1+(1.579-1)(0.7)} = \frac{1.1053}{1.4053} = 0.786$$

$$y_9 = \frac{1.579(0.8)}{1+(1.579-1)(0.8)} = \frac{1.2632}{1.4632} = 0.8633$$

$$y_{10} = \frac{1.579(0.9)}{1+(1.579-1)(0.9)} = \frac{1.4211}{1.5211} = 0.934$$

$$y_{11} = \frac{1.579(1)}{1+(1.579-1)(1)} = \frac{1.579}{1.579} = 1$$

5.

$$x = \frac{P_t - P_B^\circ}{P_A^\circ - P_B^\circ}$$

$$x_1 = \frac{760 - 333}{760 - 333} = 1$$

$$x_2 = \frac{760 - 417}{940 - 417} = 0.655$$

$$x_3 = \frac{760 - 484}{1050 - 484} = 0.487$$

$$x_4 = \frac{760 - 650}{1350 - 650} = 0.157$$

$$x_5 = \frac{760 - 760}{1540 - 760} = 0$$

$$y = \frac{P_A^\circ (x)}{Pt}$$

$$y_1 = \frac{760(1)}{760} = 1$$

$$y_2 = \frac{940(0.655)}{760} = 0.810$$

$$y_3 = \frac{1050(0.487)}{760} = 0.674$$

$$y_4 = \frac{1350(0.157)}{760} = 0.279$$

$$y_5 = \frac{1540(0)}{760} = 0$$

$$\alpha = \frac{P_A^\circ}{P_B^\circ} =$$

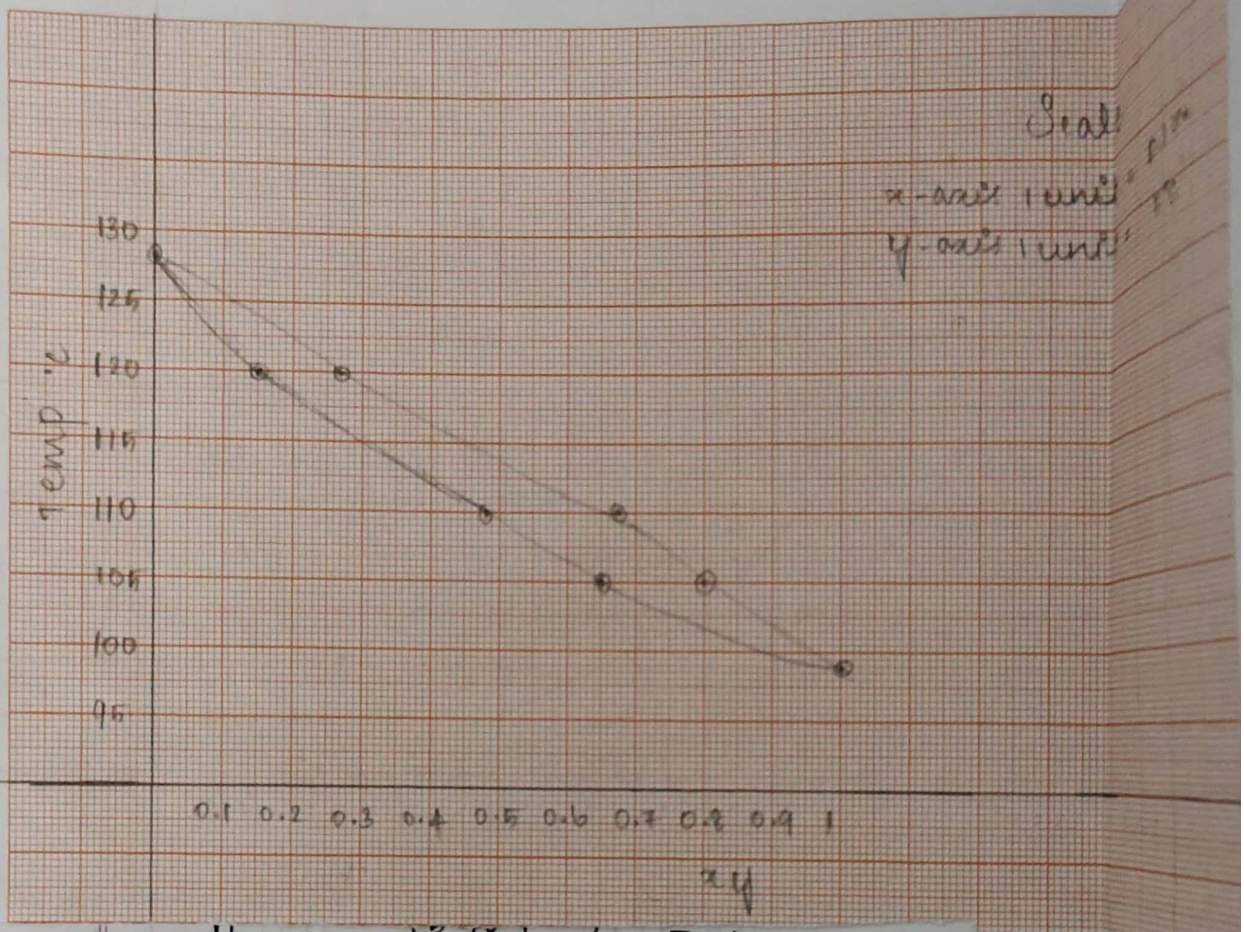
$$\alpha_1 = \frac{760}{333} = 2.28$$

$$\alpha_2 = \frac{940}{417} = 2.25$$

$$x_3 = \frac{760 - 484}{1050 - 484} = 0.487$$

$$x_4 = \frac{760 - 650}{1350 - 650} = 0.157$$

$$x_5 = \frac{760 - 760}{} = 0$$



$$y_5 = \frac{\quad}{760} = 0$$

$$\alpha = \frac{P_A^\circ}{P_B^\circ} =$$

$$\alpha_1 = \frac{760}{333} = 2.28$$

$$\alpha_2 = \frac{940}{417} = 2.25$$

$$\alpha_3 = \frac{1050}{484} = 2.17$$

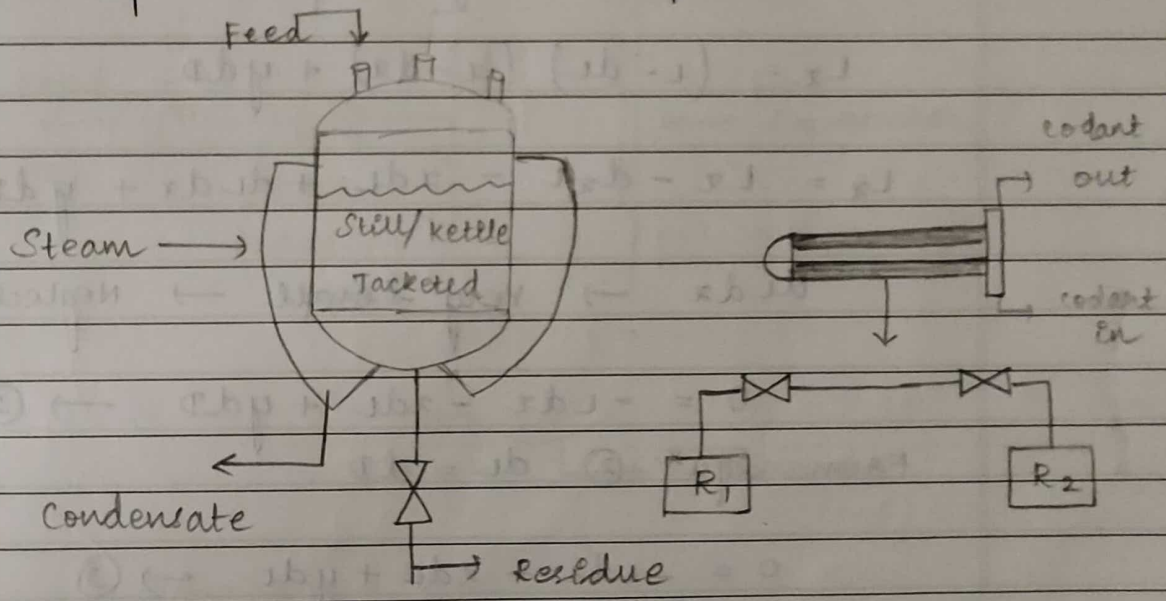
$$\alpha_4 = \frac{1250}{650} = 2.08$$

$$\alpha_5 = \frac{1540}{760} = 2.03$$

$$\alpha_{avg} = 2.162$$

8/9/22

i) Simple distillation: (batch process)

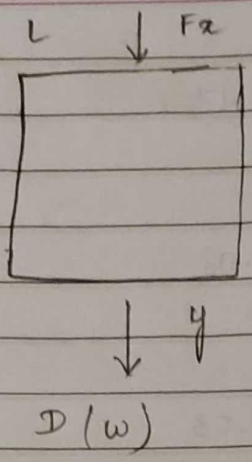


x is the mole fraction of component A in liq

y " in vapour

D → distillate

L → Amt of feed in the liq to withdraw



Overall material Balance at anytime T.

$$V = V - dL + dD \longrightarrow (1)$$

$$dL = dD$$

Material Balance of component 'A'

$$Lx = (L - dL)(x - dx) + y\,dD$$

$$Lx = Lx - dx\,L - x\,dL + dL\,dx + y\,dD$$

$dL\,dx \rightarrow$ very small $\rightarrow$ Neglected

$$0 = -L\,dx - x\,dL + y\,dD \longrightarrow (2)$$

From equ^n (2) $dL = dD$

$$0 = -L\,dx - x\,dL + y\,dL \longrightarrow (3)$$

$$0 = (y - x)\,dL - L\,dx$$

$$L.dx = (y - x)\,dL$$

$$\frac{dx}{(y - x)} = \frac{dL}{L}$$

Integrate $x \longrightarrow x_F$ to x_W

$L \Rightarrow F$ to W

$x_F \rightarrow$ Initial liq composition

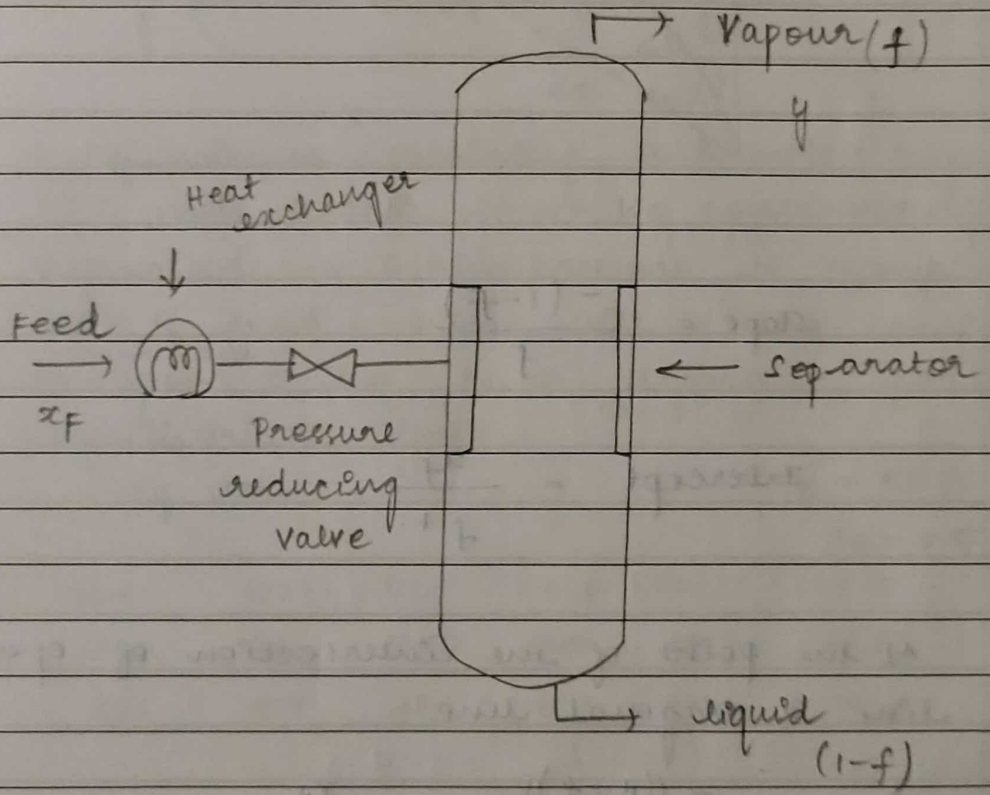
$x_w \rightarrow$ Final liq "

$$\int_{x_F}^{x_w} \frac{dx}{y-x} = \int_{F}^{W} \frac{dL}{L} \longrightarrow (4)$$

$$\ln\left[\frac{F}{W}\right] = \int_{x_F}^{x_w} \frac{dx}{(y-x)} \longrightarrow (5)$$

Eqn (5) Rayleigh's equn.

Flash (or) continuous (or) Extractive distillation:



Overall M.B at anytime 't'

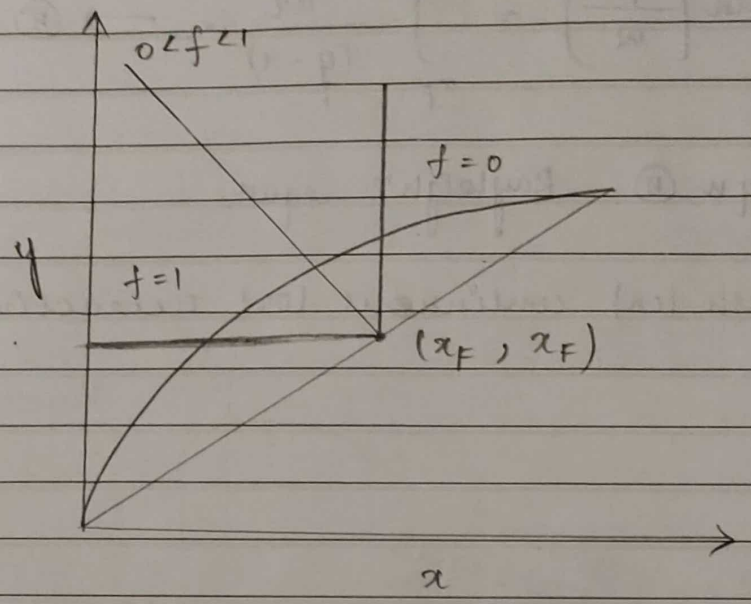
$$x_F = fy + (1-f)x \longrightarrow (1)$$

$$x_F - (1-f)x = f \cdot y$$

$$y = \frac{x_F}{f} - \frac{(1-f)x}{f}$$

$$y = -\frac{(1-f)}{f}x + \frac{x_F}{f}$$

$y = mx + c$



$$\text{slope} = \frac{-(1-f)}{f}$$

$$\text{Intercept} = \frac{x_f}{f}$$

At the point of the intersection of operating line & diagonal line

$$x = \frac{-(1-f)}{f}x + \frac{x_f}{f}$$

$$xf = -x + fx + x_F$$

$$x = x_F$$

$$x = y = x_F$$

$$\text{slope} = \frac{-(1-f)}{f} = \frac{dy}{dx}$$

$$\text{Intercept} = \frac{x_F}{f}$$

i) $f = 1$, slope

the feed is completely vapourised, Hence, the operating line is $\parallel$ le to x-axis through the point x_F on diagonal line.

No feed is vapourised.

since the operating line

F (Total Feed)

$x_F = F - D$

$= 100 - 50$

$= 50\%$.

$x_F = 40\%$.

$x_F = 0.5$

6. A liq mixture containing 50% mole n-heptane 50% mole octane is to be continuously flash vapourised at 1 atm pressure to vapourise 60% mole of feed. What will be the composition of vapour & liq in a seperator for an equilibrium stalk.

Flash distillation

60%

liq distilled

Eq. data: $x_F = 50\%$.

$x_F = 0.5$

T (°C)	98.5	105	110	115	120	125.5
P_A (mmHg)	760	940	1050	1200	1350	1540
P_B (mmHg)	333	417	484	561	650	760

$P_t = 1 \text{ atm} = 760 \text{ mmHg}$

$$x = \frac{P_t - P_B^\circ}{P_A^\circ - P_B^\circ}$$

$$x_1 = \frac{760 - 333}{760 - 333} = 1$$

$$x_2 = \frac{760 - 417}{940 - 417} = \frac{343}{523} = 0.655$$

$$x_3 = \frac{760 - 484}{1050 - 484} = \frac{276}{566} = 0.487$$

$$x_4 = \frac{760 - 561}{1200 - 561} = \frac{199}{639} = 0.311$$

$$x_5 = \frac{760 - 650}{1350 - 650} = \frac{110}{700} = 0.157$$

$$x_6 = \frac{760 - 760}{1540 - 760} = 0$$

$$y = \frac{P_A^\circ x}{P_t}$$

$$y_1 = \frac{760(1)}{760} = 1$$

$$y_2 = \frac{940(0.655)}{760} = \frac{615.7}{760} = 0.810$$

$$y_3 = \frac{1050(0.487)}{760} = 0.6728$$

$$y_4 = \frac{1200(0.311)}{760} = 0.491$$

$$y_5 = \frac{1350(0.157)}{760} = 0.278$$

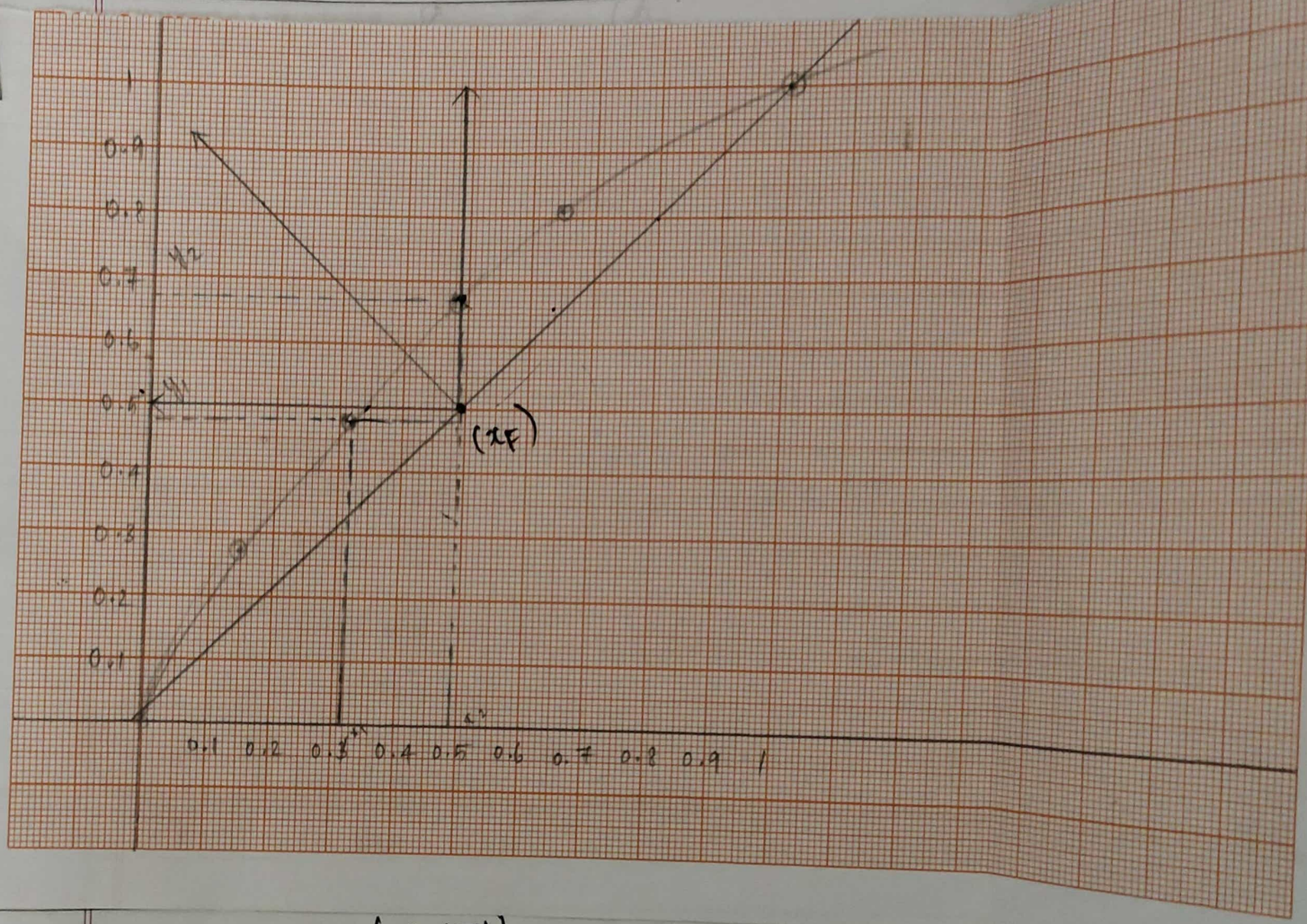
$$y_6 = \frac{1540(0)}{760} = 0$$

x	1	0.655	0.487	0.311	0.157	0
y	1	0.810	0.6728	0.491	0.278	0

$$x_3 = \frac{760 - 484}{1050 - 484} = \frac{276}{566} = 0.487$$

$$x_4 = \frac{760 - 561}{1200 - 561} = \frac{199}{639} = 0.311$$

$$x_5 = \frac{760 - 650}{1350 - 650} = \frac{110}{700} = 0.157$$

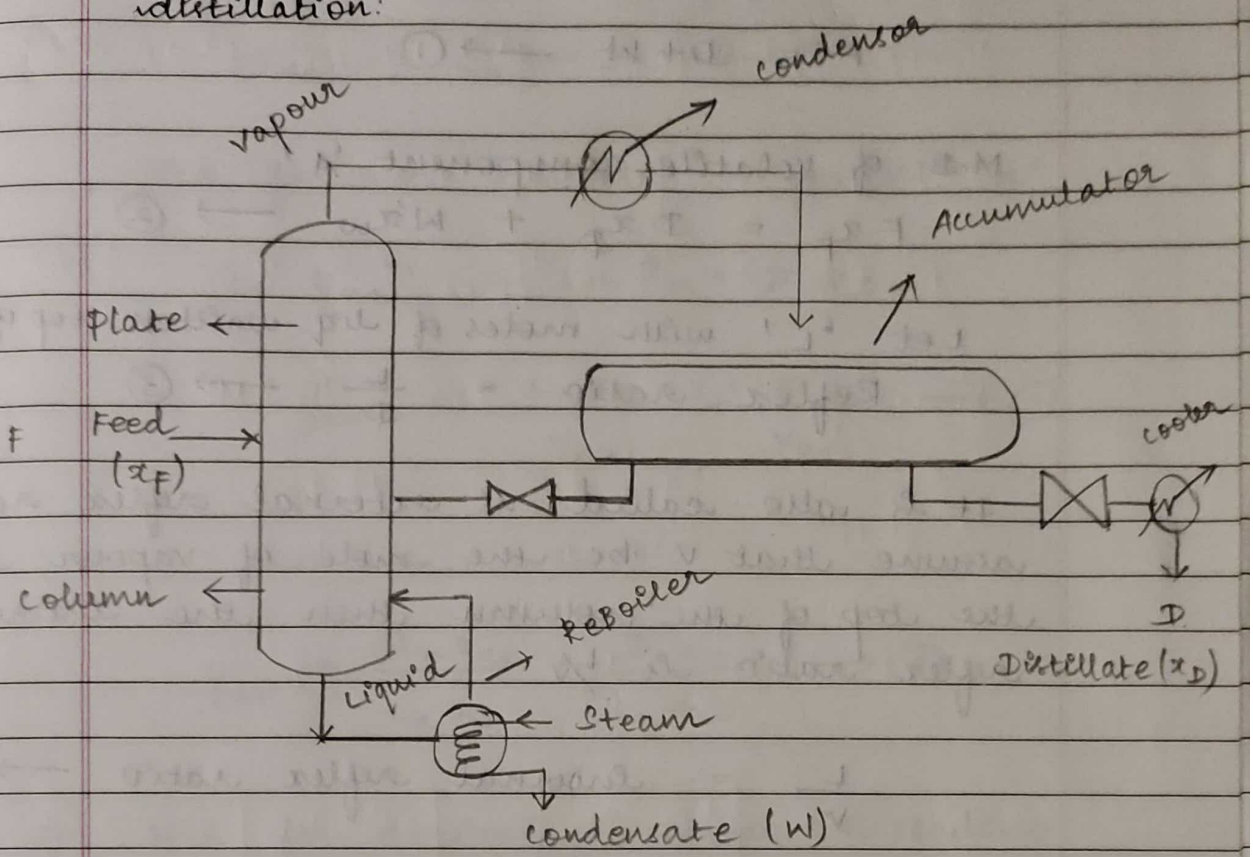


$$y_5 = \frac{1350(0.157)}{760} = 0.278$$

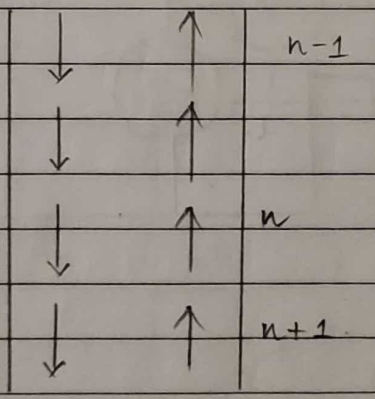
$$y_6 = \frac{1540(0)}{760} = 0$$

x	1	0.655	0.487	0.311	0.157	0
y	1	0.810	0.6728	0.491	0.278	0

Continuous Rectification (or) Fractionation distillation:



Material Balance :



F ⟶ Feed D ⟶ Distillate

W ⟶ bottom product

x_D ⟶

x_F ⟶

Overall M.B

$$F = D + W \longrightarrow (1)$$

M.B of volatile component 'A'

$$F x_F = D x_D + W x_W \longrightarrow (2)$$

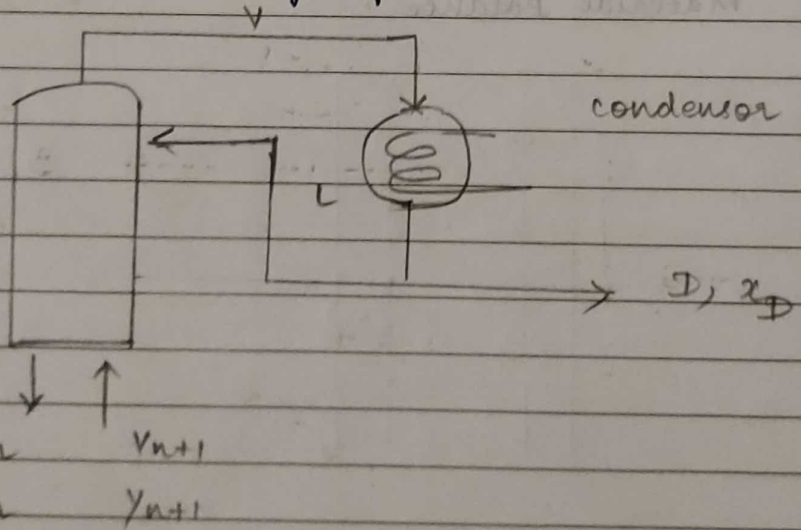
Let 'L' with moles of liq written top of the column

$$\text{Reflex ratio} = \frac{L}{D} \longrightarrow (3)$$

It is also called as external reflux ratio assume that V be the mole of vapour leaves the top of the column then the internal reflux ratio is L/V

$$\frac{L}{V} = \text{Internal reflex ratio} \longrightarrow (4)$$

M.B for rectifying section:



Overall MB for R.S

$$V = L + D \longrightarrow (1)$$

$$L = V - D$$

$$D = V - L$$

Overall M.B at n^{th} stage

$$V_{n+1} = L_n + D \longrightarrow (2)$$

M.B for more volatile component 'n'

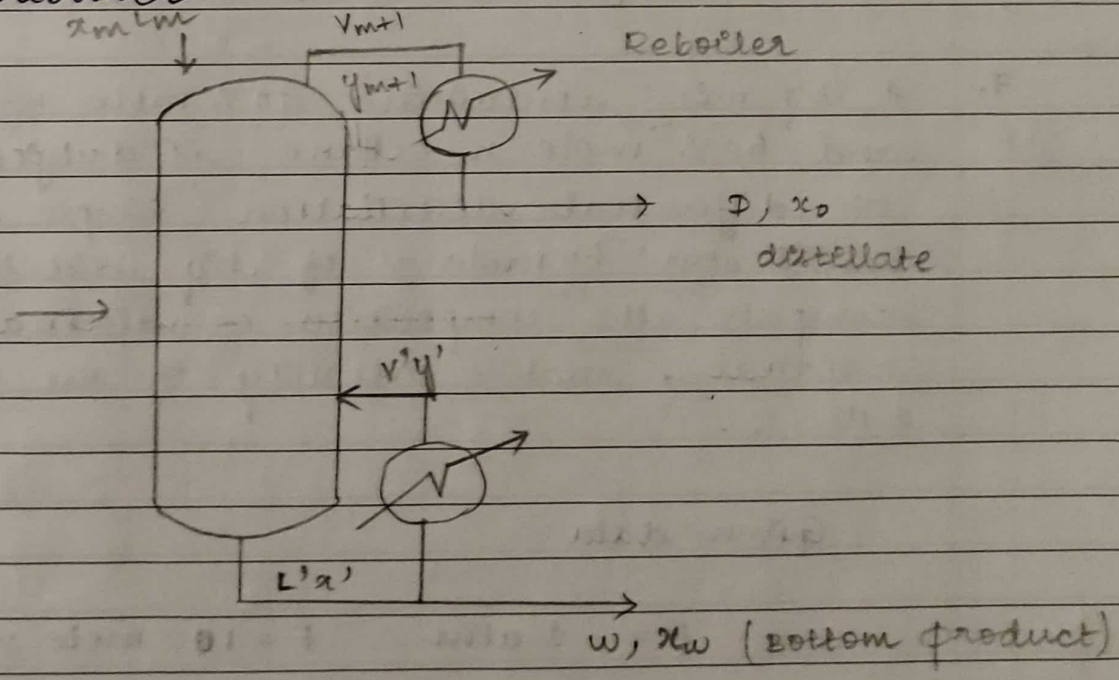
$$V_{n+1}\, y_{n+1} = L_n x_n + D \cdot x_D$$

$$y_{n+1} = \frac{L_n x_n}{V_{n+1}} + \frac{D\, x_D}{V_{n+1}} \longrightarrow (3)$$

Sub $V_{n+1} = L_n + D$

$$y_{n+1} = \frac{L_n x_n}{L_n x_D} + \frac{D\, x_D}{L_n + D} \longrightarrow (4)$$

3) M.B for stripping sector including reboiler:



over all M.B

$$L_m = V_{m+1} + W \rightarrow V_{m+1} = L_m - W$$

W → Bottom product

L_m → Liquid flow to plate $m+1$

V_{m+1} → Vapour flow from plate $m+1$

M.B for component 'A'

$$L_m x_m = V_{m+1} y_{m+1} + W \cdot x_w \longrightarrow (6)$$

$$V_{m+1} \cdot y_{m+1} = L_m \cdot x_m - W x_w$$

$$y_{m+1} = \frac{L_m \cdot x_m}{V_{m+1}} - \frac{W \cdot x_w}{V_{m+1}} \longrightarrow (7)$$

Sub $V_{m+1} = L_m - W$

$$y_{m+1} = \left[\frac{L_m x_m}{L_m - W}\right] - \left[\frac{W \cdot x_w}{L_m - W}\right] \longrightarrow (8)$$

7. A liq mix containing 40% mole n-heptane and 60% mole n-octane is subjected to differential distillation (simple distillation) at 1 atm with 60 mole % of liq distilled. Compute the composition of distillate and residual, relative volatility of the system 2.16

Given data:

$P = 1$ atm $\quad F = 100$ mole %

$D = 60$ moles $\quad x_F = 0.4$

$W = 40$ moles, $\alpha = 2.16$

Soln: Rayleigh equn:

$F = 100\,m$

$$\ln\left(\frac{F}{W}\right) = \int_{x_W}^{x_F} \frac{dx}{(y-x)}$$

$$\ln\left(\frac{F}{W}\right) = \ln\left(\frac{100}{40}\right)$$

y

$\frac{1}{y-x}$

$x =$

$y_1 =$

$y_2 =$

$y_3 =$

$y_4 =$

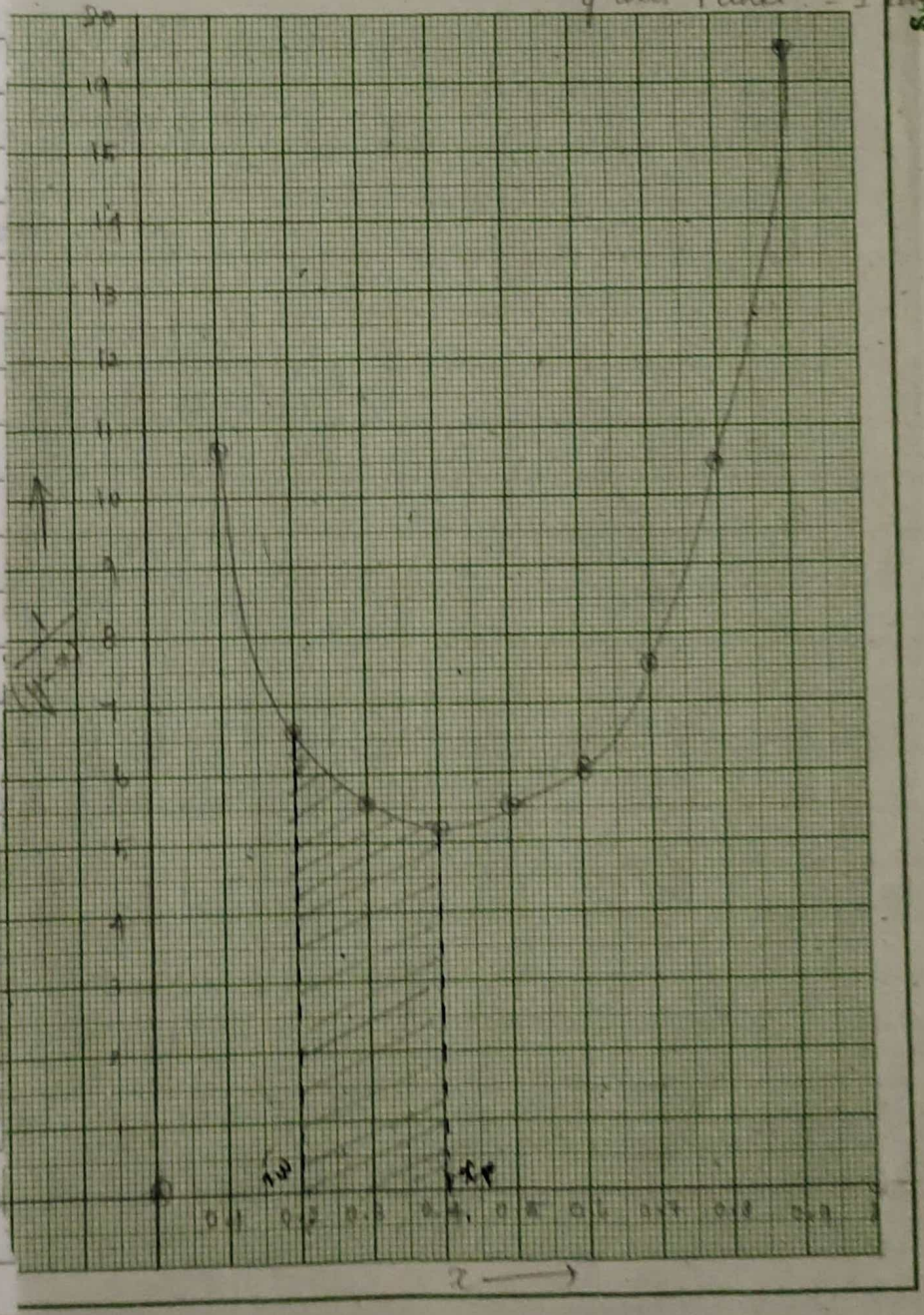
$y_5 =$

$y_6 =$

Scale

x axis 1 unit = 0.1 cm

y axis 1 unit = 1 cm



$F = 100\,m$

$$\ln\left(\frac{F}{W}\right) = \int_{x_0}^{x_F} \frac{dx}{(y-x)}$$

$$\ln\left(\frac{F}{W}\right) = \ln\left(\frac{100}{40}\right)$$

$$\ln\left(\frac{100}{40}\right) = 0.916$$

$\alpha = 2.16$

$$y = \frac{\alpha x}{1+(\alpha-1)x}$$

$\frac{1}{y-x}$

$x = 0$ to 1

$$y_1 = \frac{2.16(0)}{(1+(2.16-1)(0))} = 0$$

$$y_2 = \frac{2.16(0.1)}{(1+(2.16-1)(0.1)} = \frac{0.216}{1.116} = 0.193$$

$$y_3 = \frac{2.16(0.2)}{(1+(2.16-1)(0.2))} = \frac{0.432}{1.232} = 0.350$$

$$y_4 = \frac{2.16(0.3)}{(1+(2.16-1)(0.3))} = \frac{0.648}{1.348} = 0.480$$

$$y_5 = \frac{2.16(0.4)}{(1+(2.16-1)(0.4)} = \frac{0.864}{1.464} = 0.587$$

$$y_6 = \frac{2.16(0.5)}{(1+(2.16-1)(0.5))} = \frac{1.08}{1.58} = 0.683$$

$$y_7 = \frac{2.16(0.6)}{(1+(2.16-1)(0.6))} = \frac{1.296}{1.696} = 0.764$$

$\frac{1}{(y-x)}$

0
10.75
6.66
5.52
5.26
5.46
7.09
7.46
10.4
19.6
0

$$y_8 = \frac{2.16(0.7)}{(1+(2.16-1)(0.7))} = \frac{1.512}{1.812} = 0.834$$

$$y_9 = \frac{2.16(0.8)}{(1+(2.16-1)(0.8))} = \frac{1.728}{1.928} = 0.896$$

$$y_{10} = \frac{2.16(0.9)}{(1+(2.16-1)(0.9))} = \frac{1.944}{2.044} = 0.951$$

$$y_{11} = \frac{2.16(1)}{(1+(2.16-1)(1))} = 1$$

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8. A liq mix containing 35% mole n-heptane and 65% mole n-octane is subjected to differential distillation at 1 atm with 65 mole % of liq distilled. compute the composition of composited distillate and residual, heptane is more volatality, relative volatality of the system 2.16.

$P = 1$

$W = 35\%$ $\quad D = 65\%$

$F = 35$ $\quad x_F = 0.35$

$\alpha = 2.16$

Rayleigh Equation

$$\ln\left(\frac{F}{W}\right) = \int_{xw}^{xf} \frac{dx}{(y-x)}$$

$$\ln\left(\frac{100}{35}\right) = 1.049$$

$$y = \frac{\alpha x}{1+(\alpha-1)x} \qquad x \rightarrow 0 \text{ to } 1$$

x	y	$1/(y-x)$
0	0	0
0.1	0.193	10.75
0.2	0.350	6.66
0.3	0.480	5.52
0.4	0.587	5.26
0.5	0.683	5.46
0.6	0.764	6.09
0.7	0.834	7.46
0.8	0.896	10.4
0.9	0.951	19.6
1	1	0

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7.

TO find area:

$$(\text{No. of. boxes}) \times \left(\begin{matrix}x\text{-axis}\\ \text{scale}\end{matrix}\right) \times \left(\begin{matrix}y\text{-axis}\\ \text{scale}\end{matrix}\right) = 0.916$$

$$(\text{No. of. boxes})\ (0.1)\ (1) = 0.916$$

$$\text{No. of. boxes} = \frac{0.916}{0.1}$$

$$= 9.16$$

$F = W + D$

$x_F F = W x_W + D y_D$

$F \cdot x_F = W x_W + D y_D$

$F = 100, \quad W = 40, \quad D = 60$

$(100)(0.4) = 40(0.2) + 60\, y_D$

$y_D = 0.53$

$x_W = 0.2 \qquad x_F = 0.4$

Result:

composition of composited Destillate 0.53 mole
Residue = 0.2 mole fract
of heptan

8.

No. of. boxes $\times$ (x axis scale) $\times$ (y-axis scale) = 1.049

No. of. boxes $\times$ 0.1 $\times$ 1 = 1.049

2

$$\text{No. of. boxes} = \frac{1.049}{0.1}$$

$$= 10.49$$

$x_W = 0.16 \qquad x_F = 0.35$

$F \cdot x_F = W \cdot x_W + D y_D$

$F = 100, \quad W = 35, \quad D = 65$

$100(0.35) = 35(0.16) + 65(y_D)$

$$y_D = \frac{35 - 5.6}{65} = \frac{29.4}{65}$$

$$= 0.45$$

Stepwise procedure by obtaining theoratical plates:

Step 1: Material Balance, Evaluate the terms D, W, L etc...

Step 2: Draw the equilibrium curve and diagonal with the help of x-y data given

If relative volatality is given, generate x-y data using the relative value provided

Step 3: Draw the operating line of rectifying section through the point (x_D, x_D) on the diagonal and with an intercept on the diagonal and with an intercept equal to $\frac{x_D}{R+1}$

$$\frac{Dx_D}{L \text{ or } D} \text{ of slope} = \frac{R}{R+1}$$

Step 4: Draw the operating line of stripping section through the point

(x_w, x_w) on the diagonal with a slope of $\left[\frac{L'}{L'-W}\right]$

Step 5: Starting from the point (x_D, x_D) on diagonal, draw a horizontal line to meet the equilibrium curve at point-1.

Draw a vertical point A to meet operating line from point 1.

Step 6: proceed in this way, i.e., constructing a triangle b/w equilibrium curve and operating line of rectifying section. Still we are above the point of intersection of the two operating line, once there point of intersection construct a Δle b/w equilibrium curve and operating line of stripping section i.e., dropping verticals on operating line of stripping section.

Step 7: proceed in the same Manner still we reach / across the point (x_w, x_w)

Step 8: count the triangle constructed b/w x_D and x_w. Each triangle on the equilibrium diagram represent a theoretical plate.

Step 9: If no. of triangle (n) then n represent the theoratical, no. of theoratical plates including reboiler and n-1 represent

The No. of. theoratical plate in a column, since the vapour and liq. leaving reboiler or in equilibrium, the reboiler is equal to 1 theoratical plates.

Limitations of Mecabbe - Theiele Method:

It is not used when the relative volatility is less than 1.3 or greater than 5.

It is not used when more than 25 theoratical plates or stages required.

It is not used when the operating reflex ratio is less than 1.1 times, the Min reflex ratio.

Question:

1. 10 mole % of benzene, the feed enters a column and its bubble point it is proposed to the operate the column with reflex ratio of 2.5 it is required to find the no. of theoratical plates needed and position of feed plate. The vapour liq equilibrium data are given below.

x	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	0	0.13	0.21	0.375	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1

A MB of 1 wt % of the feed 10 mole % b it to be seperated to give a product of 95 % ben & bottom product

A mixture of benzene & toluene containing 40 & 60 wt % toluene is to be seperated by fractionating column to give a 96 % of wt of benzene & the bottom containing 95 wt % of toluene. The Feed is the mixture of

$\frac{2}{3}^{rd}$ vapour & $\frac{1}{3}^{rd}$ liq. Find out the no. of theoratical stages required. If the reflux ratio of 1.5 times the min to be used & the relative volatality is 2.5 times

3. A mixture of benzene & toluene containing 40 mole %. to be benzene and is to be seperated to give a 90 mole %. benzene at a top & a bottom with not more than 10%. benzene using an avg. relative volatality of 2.4. calculate the No. of. theoratical plates required at total reflux also calcula the minimal reflux ratio and the feed is liq. and its bubble point.

4. A rectification column is feed with 100 kmo of mix. containing 50%. mole of hexane & 50 mole %. of Octane & its absolute pressure is 101.325 kpa. The feed that its Boiling point, the distillate contains 90 mole %. hexane & the bottom contains 10 mole %. of hexane. The reflux ratio is 3:1. calculate kmol/hr distillate & kmol/hr bottoms & the no. of. theoratical plates.

x	1	0.69	0.4	0.192	0.045	0
y	1	0.932	0.78	0.538	0.1775	0

5. 1000 kmol of an ethanol propanol mix. containin 65 mole %. of ethanol is to be seperated in a continuous plate column. operating at 101.32 kpa. The desired terminal composition in terms of mole frac. of ethanol are $x_D = 0.92$ & $x_W = 0.07$ The feed is saturated vapour & total condensor is used. When the reflex flow rate is 4 time the amt of top product, find the No. of. theoratical plates required for seperation. The relative volatality of ethanol

Divide ③ / ④

$$\frac{L_n}{V_n} = \frac{L_{N+1} + S_{N-1} - S_N}{V_{N+1} + W_{N-1} - W_N} = x_n$$

$x = L/V$

$$x_n = \frac{L_n}{V_n} = \frac{L_{N+1} + S_{N-1} - S_N}{V_{N+1} + W_{N-1} - W_N}$$

$x_n = L/V_n$

↳ ⑤

$W = a + bx \longrightarrow ⑥$

$S = xW \longrightarrow ⑦$

$$x = \frac{\text{Kg solute}}{\text{Kg soln}} \Rightarrow W = \frac{\text{Kg soln}}{\text{Kg insoluble solid}}$$

19/10/22 UNIT - 3:

Mcabbean's Method: [calculation of q value]

Assume Feed is 'F'

For stripping section:

$L' = L + qF$

For rectifying section

$V = V' + (1-q)F$

Energy to convert 1 Mole

$$q = \frac{\text{To Saturated vapour}}{\text{Molal latent heat vapour}}$$

For q line (or) Feed line

$$y = \left[\frac{-q}{1-q}\right] x + \left[\frac{x_F}{1-q}\right] \longrightarrow ⑦$$

$y = mx + c$ slope $= -q/1-q$ $y = x \rightarrow (8)$

The point of intersection of the feed line on diagnol $y = x$

$$x = \frac{-q}{1-q}x + \frac{x_F}{1-q} \rightarrow (9)$$

$$x = \frac{1}{1-q}(-q \cdot x) + x_F$$

$$(1-q)x = -qx + x_F$$

$$x - qx = -qx + x_F$$

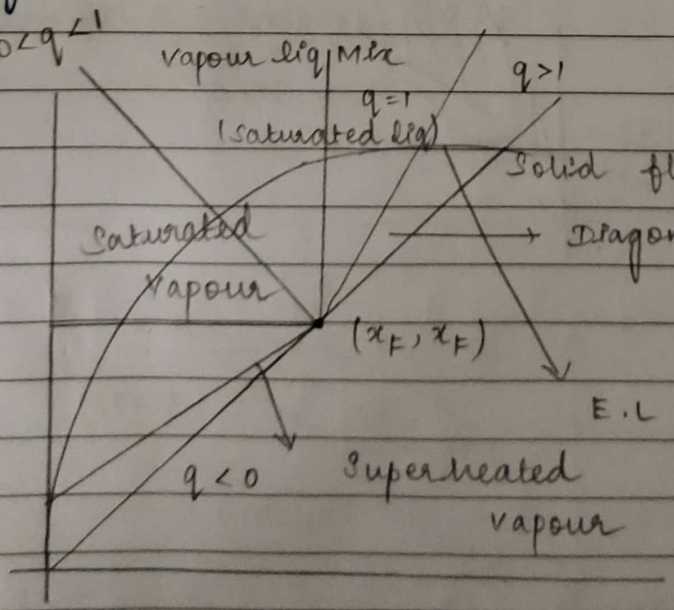
$x = x_F$; $y = x_F$ (x_F, x_F)

For $q = 1$ $\Rightarrow$ $-1/0 = \infty$

For $q = 0$ $\Rightarrow$ $-q/1-q = 0$

(x-axis)

Hence, the q line is $\parallel^{le}$ to y-axis through the point (x_F, x_F) on diagnol.



Determination of (SOL)

$$y = \frac{L'}{L'-W} \cdot x - \left(\frac{W \cdot x_W}{L'-W}\right) \longrightarrow (10)$$

at the diagonal, $x = y$

$$x = \frac{L'}{L'-W} x - \frac{W \cdot x_W}{L'-W} \longrightarrow (11)$$

$$x(L'-W) = L'x - Wx_W$$

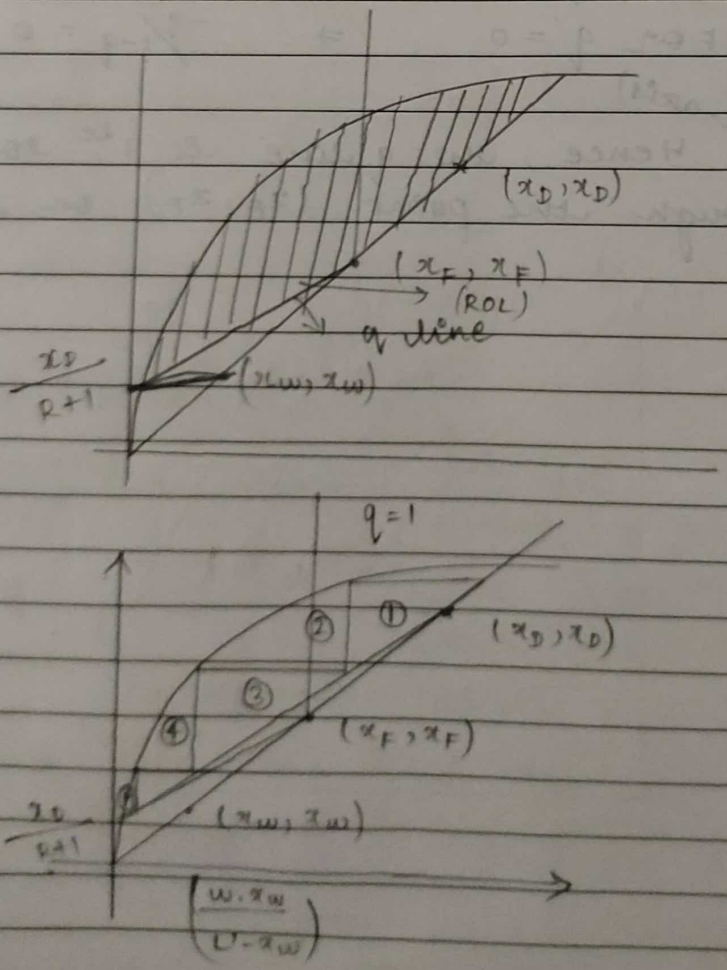
$$xL' - xW = L'x - Wx_W$$

$$x = x_W$$

$$y = x_W$$

Intersection point (x_W, x_W)

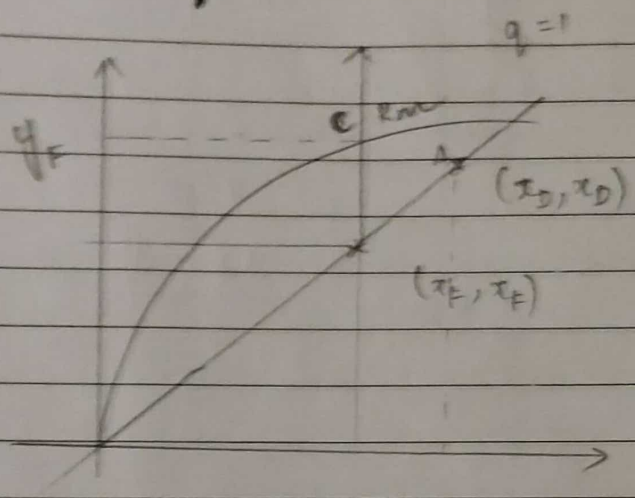
Draw the stripping section operating line through the point (x_W, x_W) on the diagonal with intercept of $\left(\frac{W \cdot x_W}{L-W}\right)$



When the operating line moves towards the diagonal, The OL Meet at the E.L that point is called Minimum reflux ratio.

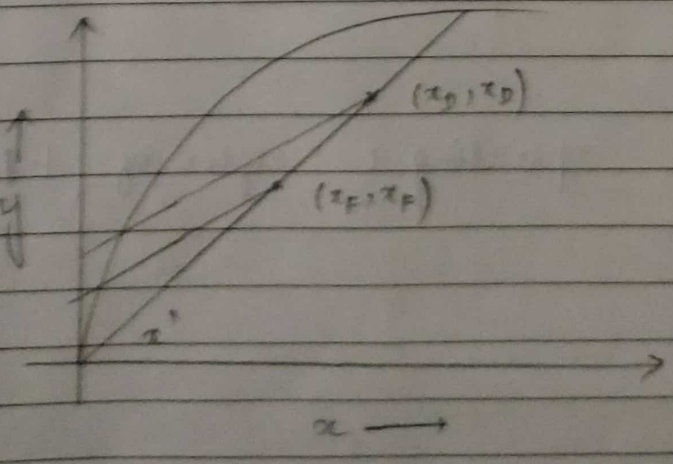
Calculation of Minimum Reflux ratio:

i) Saturated liq:



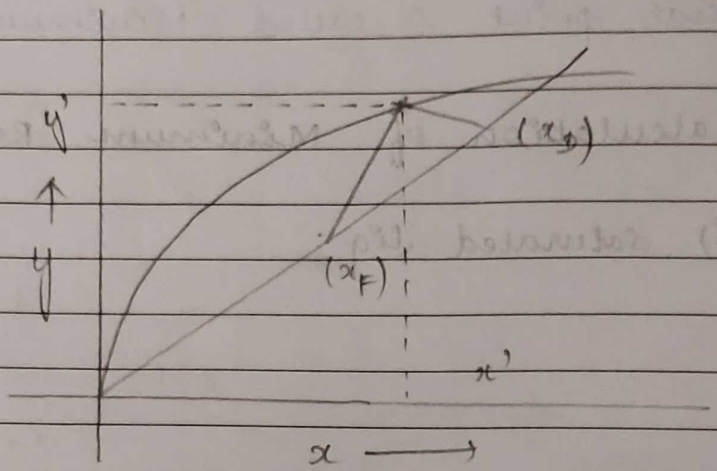
$$R_m = \frac{x_D - y_F}{y_F - x_F}$$

iii) Saturated vapour ($q=0$)



$$R_m = \frac{x_D - x_F}{x_F - x'}$$

iii) cold liquid : (q>1)



$$R_m = \frac{x_D - y'}{\quad}$$

operating reflux of 1.0 to 1.5

Answer:

1. Given data:

$x_F = 60\% = 0.6$

$x_D = 95\% = 0.95$

$x_W = 10\% = 0.01$

Reflux ratio (R) $= 2.5$

Feed condition (i.e Bubble point) = Saturated liq ($q = 1$).

Intercept for (ROL) $= \frac{x_D}{R+1} = \frac{0.95}{(2.5+1)}$

$= 0.2714$

i) No. of. Theoratical stages required is = 11 Stages.

ii)

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2.

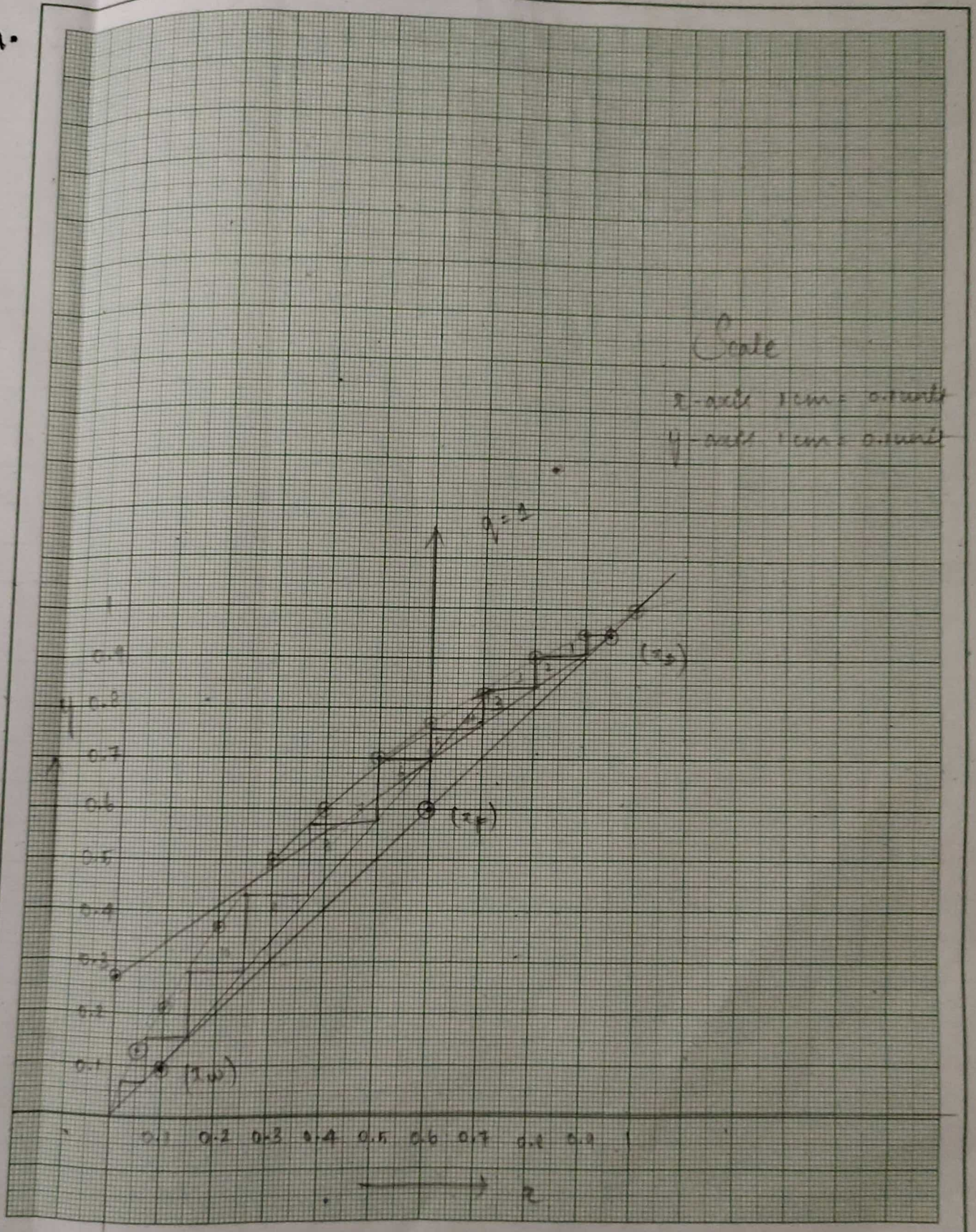
$x_{A_1} = 40\%$; $x_{A_2} = 60\%$

Mol. wt of Benzene = 78

Toluene = 92

$$x_F = \frac{\left[\frac{40}{78}\right]}{\left[\frac{40}{78}\right] + \left[\frac{60}{92}\right]} = 0.43$$

10


Scale
x-axis 1cm = 0.1 unit
y-axis 1cm = 0.1 unit
y = x
(x3)
(x4)
(x5)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
x

Shine

$$x_D = \frac{\left[\frac{96}{78}\right]}{\left[\frac{96}{78}\right] + \left[\frac{4}{92}\right]} = 0.965$$

$$x_W = \frac{\left[\frac{5}{78}\right]}{\left[\frac{5}{78}\right] + \left[\frac{95}{92}\right]} = 0.0584$$

Relative volatality (α) = 2.5

$$y = \frac{\alpha(x)}{1 + (\alpha - 1)x}$$

$$y_1 = \frac{2.5\,(0)}{1 + (2.5-1)(0)} = 0$$

$$y_2 = \frac{2.5\,(0.1)}{1 + (2.5-1)(0.1)} = \frac{0.25}{1.15} = 0.217$$

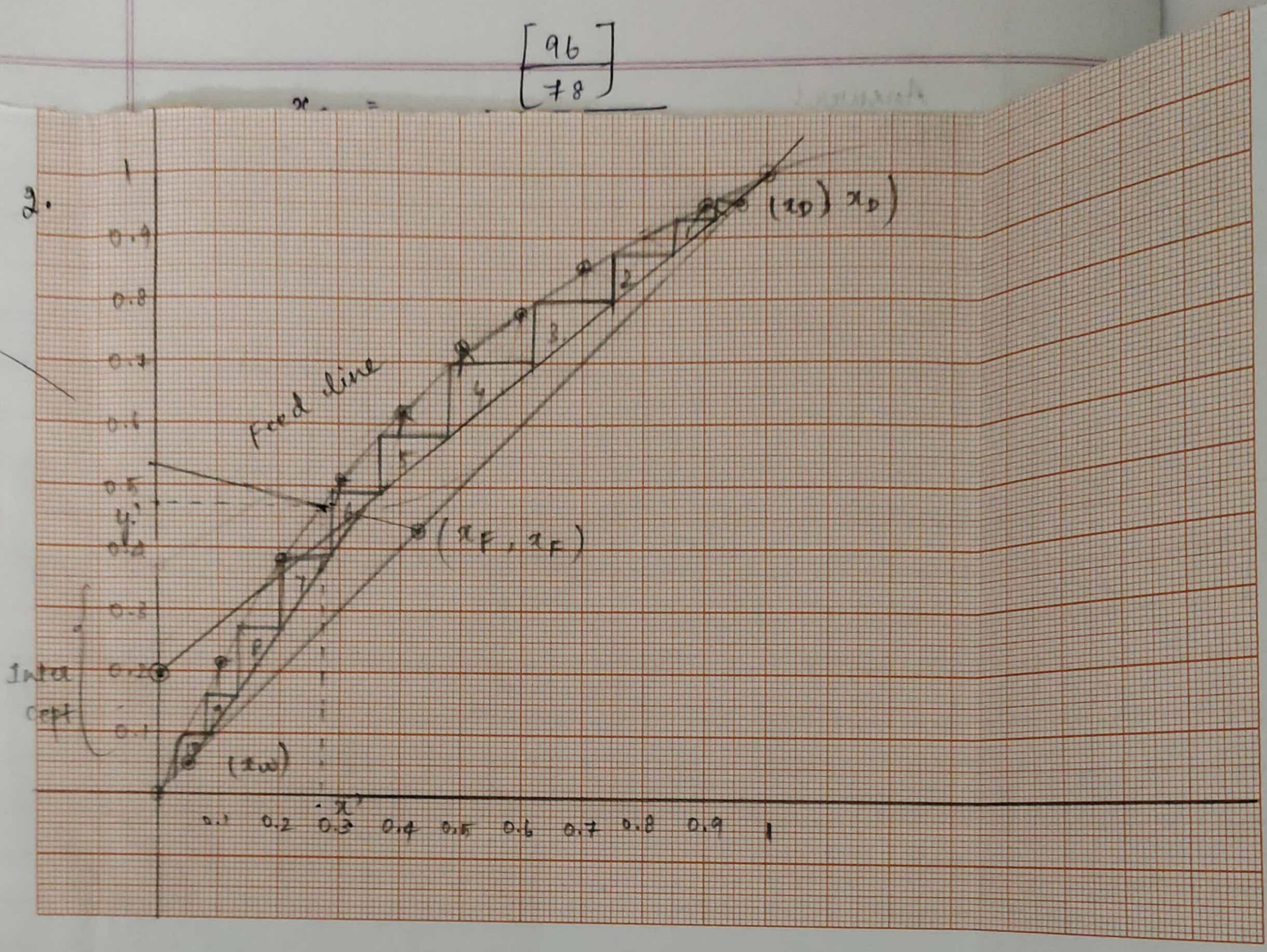
$$y_3 = \frac{2.5\,(0.2)}{1 + (2.5-1)(0.2)} = 0.384$$

$$y_4 = \frac{2.5\,(0.3)}{1 + (2.5-1)(0.3)} = 0.517$$

$$y_5 = \frac{2.5\,(0.4)}{1 + (2.5-1)(0.4)} = 0.625$$

$x_w = \frac{96}{78}$

2.



$$y_2 = \frac{2.5\,(0.1)}{1 + (2.5-1)(0.1)} = \frac{0.25}{1.15} = 0.217$$

$$y_3 = \frac{2.5(0.2)}{1 + (2.5-1)(0.2)} = 0.384$$

$$y_4 = \frac{2.5\,(0.3)}{1 + (2.5-1)(0.3)} = 0.517$$

$$y_5 = \frac{2.5\,(0.4)}{1 + (2.5-1)(0.4)} = 0.625$$

$$y_6 = \frac{2.5(0.5)}{1+(2.5-1)(0.5)} = \frac{1.25}{1.75} = 0.714$$

$$y_7 = \frac{2.5(0.6)}{1+(2.5-1)(0.6)} = \frac{1.5}{1.9} = 0.789$$

$$y_8 = \frac{2.5(0.7)}{1+(2.5-1)(0.7)} = \frac{1.75}{2.05} = 0.853$$

$$y_9 = \frac{2.5(0.8)}{1+(2.5-1)(0.8)} = \frac{2}{2.2} = 0.909$$

$$y_{10} = \frac{2.5(0.9)}{1+(2.5-1)(0.9)} = \frac{2.25}{2.35} = 0.957$$

$$y_{11} = \frac{2.5(1)}{1+(2.5-1)(1)} = 1$$

$$\frac{1}{3} - \frac{2}{3}$$

$$-(1-8)$$

$$8$$

$$q = \frac{-q}{1-q} = \frac{-1/3}{1+1/3}$$

$$q = -0.25$$

$$\theta = \tan^{-1}(-0.25)$$

$$\theta = -14.02^\circ$$

$$Rm = \frac{x_D - y'}{y' - x'} = \frac{0.965 - 0.47}{0.47 - 0.27}$$

$$= \frac{0.495}{0.2} = 2.475$$

$R = 1.5\ (R_m)$

$= 1.5\ (2.475)$

$R = 3.712$

$$\text{Intercept} = \frac{x_D}{R+1}$$

$$= \frac{0.965}{3.712+1}$$

$$= 0.204$$

3. need to draw section of stripping & rectifying

» $x_F = 40\% = 0.4$

$x_D = 90\% = 0.9$

$x_W = 10\% = 0.1$

$\alpha_{avg} = 2.4$

$$y = \frac{\alpha x}{1+(\alpha-1)\,x}$$

$$y_0 = \frac{2.4(0)}{1+(2.4-1)(0)} = 0$$

$$y_1 = \frac{2.4(0.1)}{1+(2.4-1)(0.1)} = \frac{0.24}{1.14} = 0.21$$

$$y_2 = \frac{2.4(0.2)}{1+(2.4-1)(0.2)} = \frac{0.48}{1.28} = 0.37$$

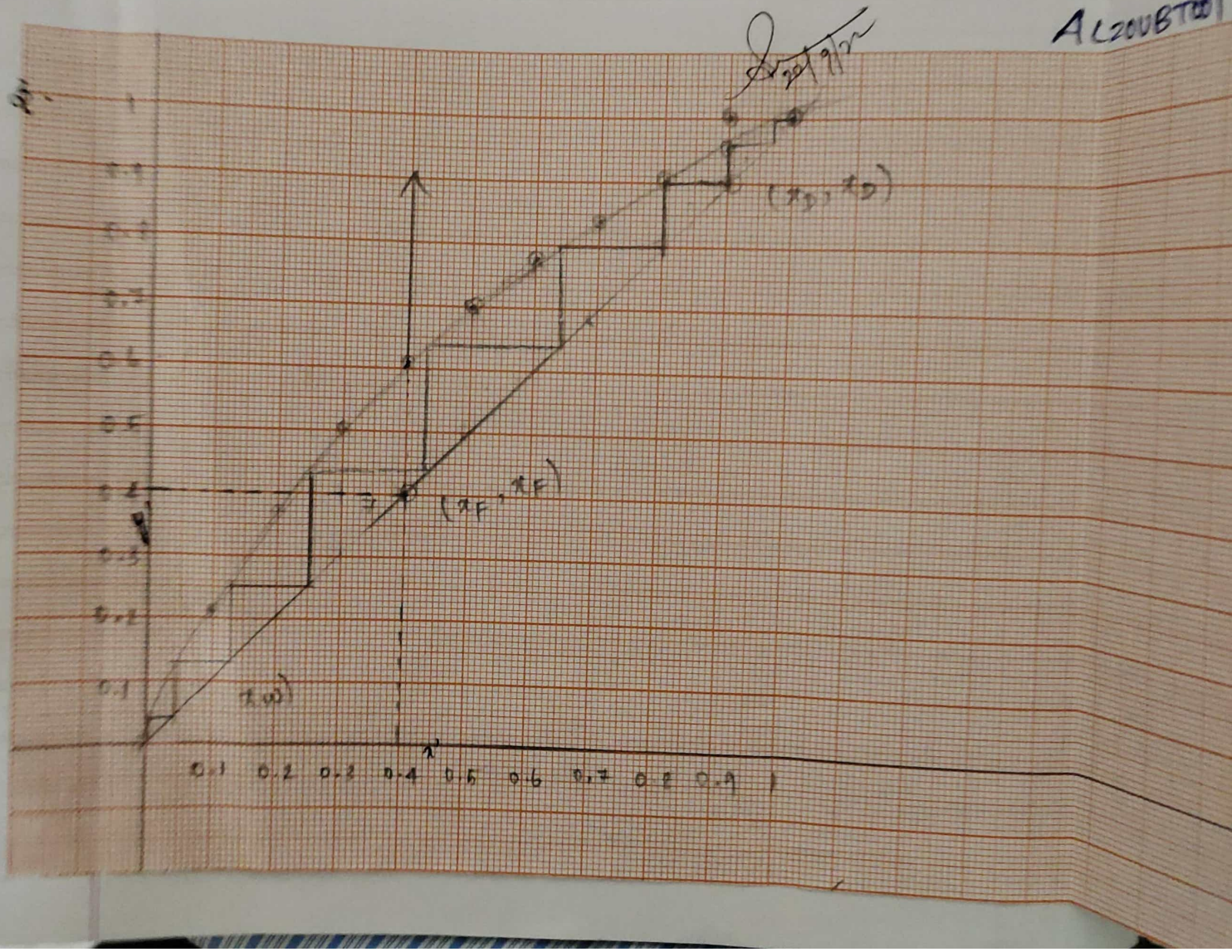
$$= \frac{0.965}{3.712 + 1}$$

$$= 0.204$$

$x_F = 40\% = 0.4$

$x_D = 90\% = 0.9$

$x_W = 10\% = 0.1$



$$y_3 = \frac{2.4(0.3)}{1+(2.4-1)(0.3)} = \frac{0.72}{1.42} = 0.50$$

$$y_4 = \frac{2.4(0.4)}{1+(2.4-1)(0.4)} = \frac{0.96}{1.56} = 0.61$$

$$y_5 = \frac{2.4(0.5)}{1+(2.4-1)(0.5)} = \frac{1.2}{1.7} = 0.70$$

$$y_6 = \frac{2.4(0.6)}{1+(2.4-1)(0.6)} = \frac{1.44}{1.84} = 0.78$$

$$y_7 = \frac{2.4(0.7)}{1+(2.4-1)(0.7)} = \frac{1.68}{1.98} = 0.84$$

$$y_8 = \frac{2.4(0.8)}{1+(2.4-1)(0.8)} = \frac{1.92}{2.12} = 0.90$$

$$y_9 = \frac{2.4(0.9)}{1+(2.4-1)(0.9)} = \frac{2.16}{2.26} = 0.95$$

$$y_{10} = \frac{2.4(1)}{1+(2.4-1)(1)} = 1$$

$$R_m = \frac{x_D - y'}{y' - x'} = \frac{0.9 - 0.61}{0.61 - 0.4}$$

$$= \frac{0.29}{0.21} = 1.38$$

x_F = 50% = 0.5

x_D = 90% = 0.9

x_W = 10% = 0.1

Reflux ratio = 3:1

R = 3

Intercept $\frac{x_D}{R+1} = \frac{0.9}{3+1} = 0.225$

x	1	0.69	0.4	0.192	0.045	0
y	1	0.932	0.78	0.53	0.17	0

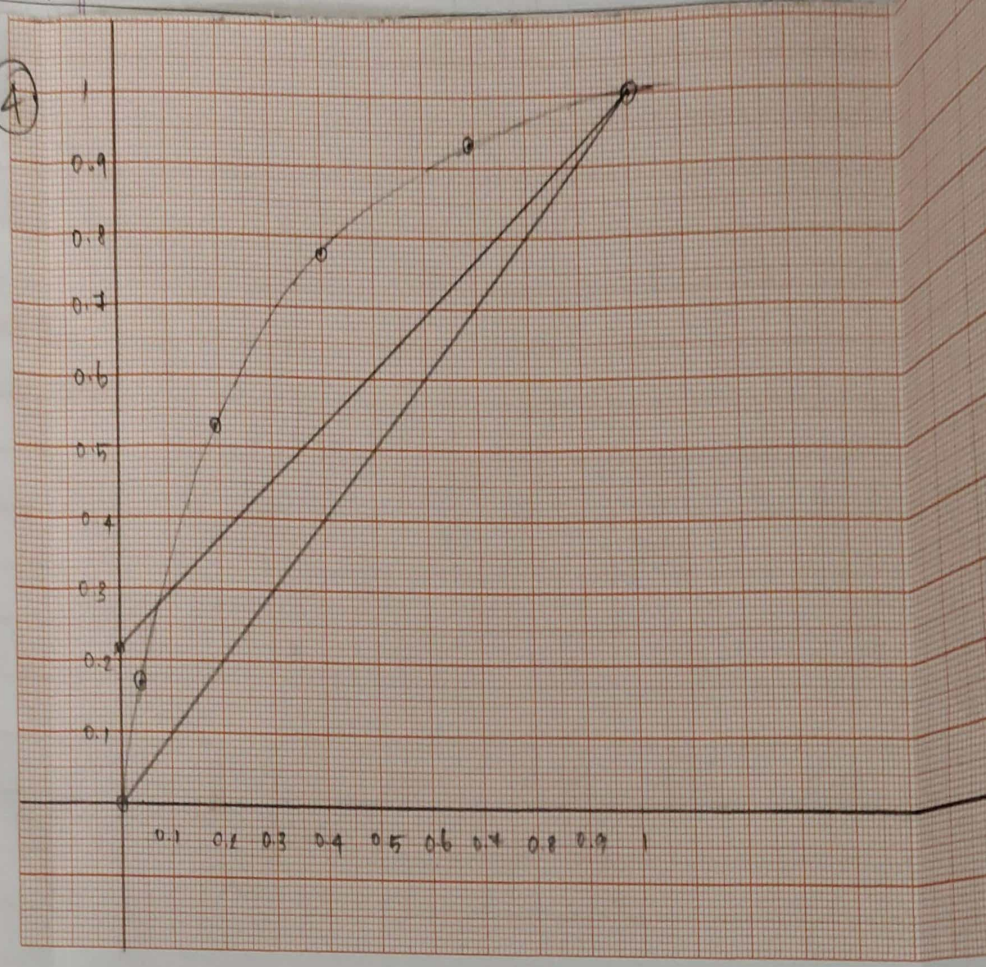
1/22

6. In equilibrium Data for the ~~boiling~~ binding system heptane and ethyl benzene are as follows a composed of 42 mole % of heptane & 58 mole % of ethyl benzene is to be fractionating to yield a overload product of 97 mole % Heptane and a residue of 99 mole % ethyl benzene. The reflux ratio 1.5 and a feed is a saturated liq and at a bubble point

i) Determine Minimum reflux ratio.

ii) Determine the no. of equilibrium stages at total reflux.

4.



binding

b. In equilibrium Data for the ~~boiling~~ system heptane and ethyl benzene are as follows a composed of 42 mole % of heptane & 58 mole % of ethyl benzene is to be fractionating to yield a overload product of 97 mole % Heptane and a residue of 99 mole % ethyl benzene. The reflux ratio 1.5 and a feed is a saturated liq and at a bubble point

i) Determine Minimum reflux ration.

ii) Determine the no. of equilibrium stages at total reflux.

x	0	0.08	0.18	0.25	0.33	0.48	0.65	0.78	0.914	1
y	0	0.23	0.42	0.51	0.60	0.72	0.83	0.9	0.963	1

Given data:

$x_F = 42\% = 0.42$

$x_D = 97\% = 0.97$

$x_W = 1\% = 0.01$

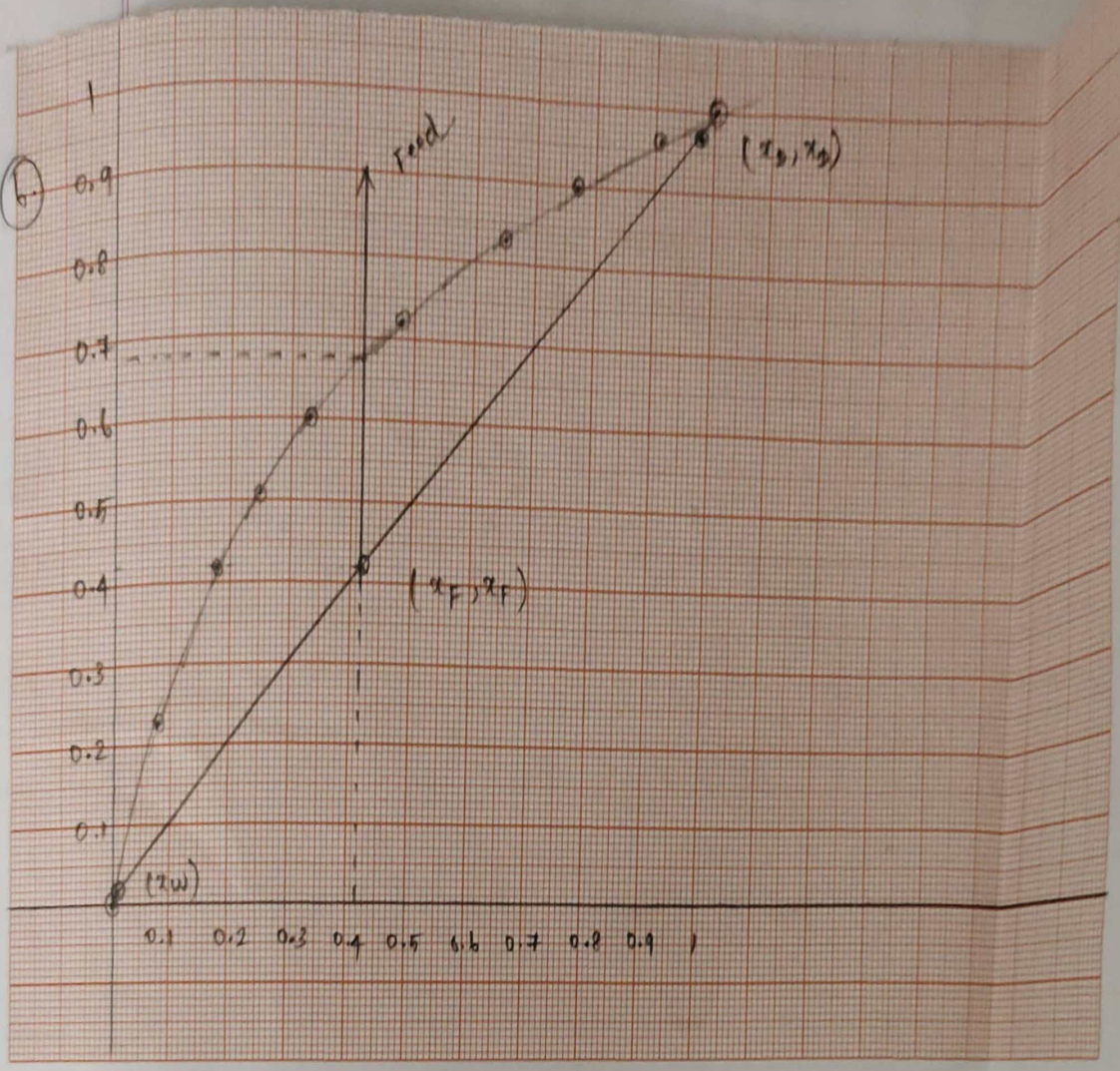
$R = 1.5$

Feed saturated liq $q = 1$ 11^{de} y-axis.

$$R_m = \frac{x_D - y_F'}{y_F' - x_F'} = \frac{0.97 - 0.67}{0.67 - 0.42}$$

$$R_m = 1.2$$

7. A bubble cap fractionating column of 12 plates working at an average efficiency of 75% is being used to distillate 1000 kg/h of aq. methanol at its bubble point enters the tower. The Feed overhead product & bottom product are 25 mole %, 90 % & 10 mole %. respectively. the



7. A bubble cap fractionating column of 12 plates working at an averaged efficiency of 75%. is being used to distillate 1000 kg/h of aq. methanol at its bubble point enters the tower. The Feed overhead product & bottom product are 25 mole %, 90 % & 10 mole %. respectively. the

x	y
0	0
0.08	0.365
0.1	0.418
0.2	0.579
0.3	0.665
0.4	0.729
0.5	0.778
0.7	0.87
0.8	0.928
0.95	0.979
1	1

x_F = 50% = 0.5

x_D = 90% = 0.9

x_W = 10% = 0.1

Feed is at bubble point

$q = 1$

$R = 1.7\ (R_m)$

$y_F' = 0.78$

$x_F' = 0.5$

$$R_m = \frac{x_D - y_F'}{y_F' - x_F'} = 0.42$$

$R = (0.42)\ 1.7 \Rightarrow 0.714$

$$\text{Intercept of ROL} = \frac{x_D}{R+1} = \frac{0.9}{[0.714+1]}$$

$$= 0.52.$$

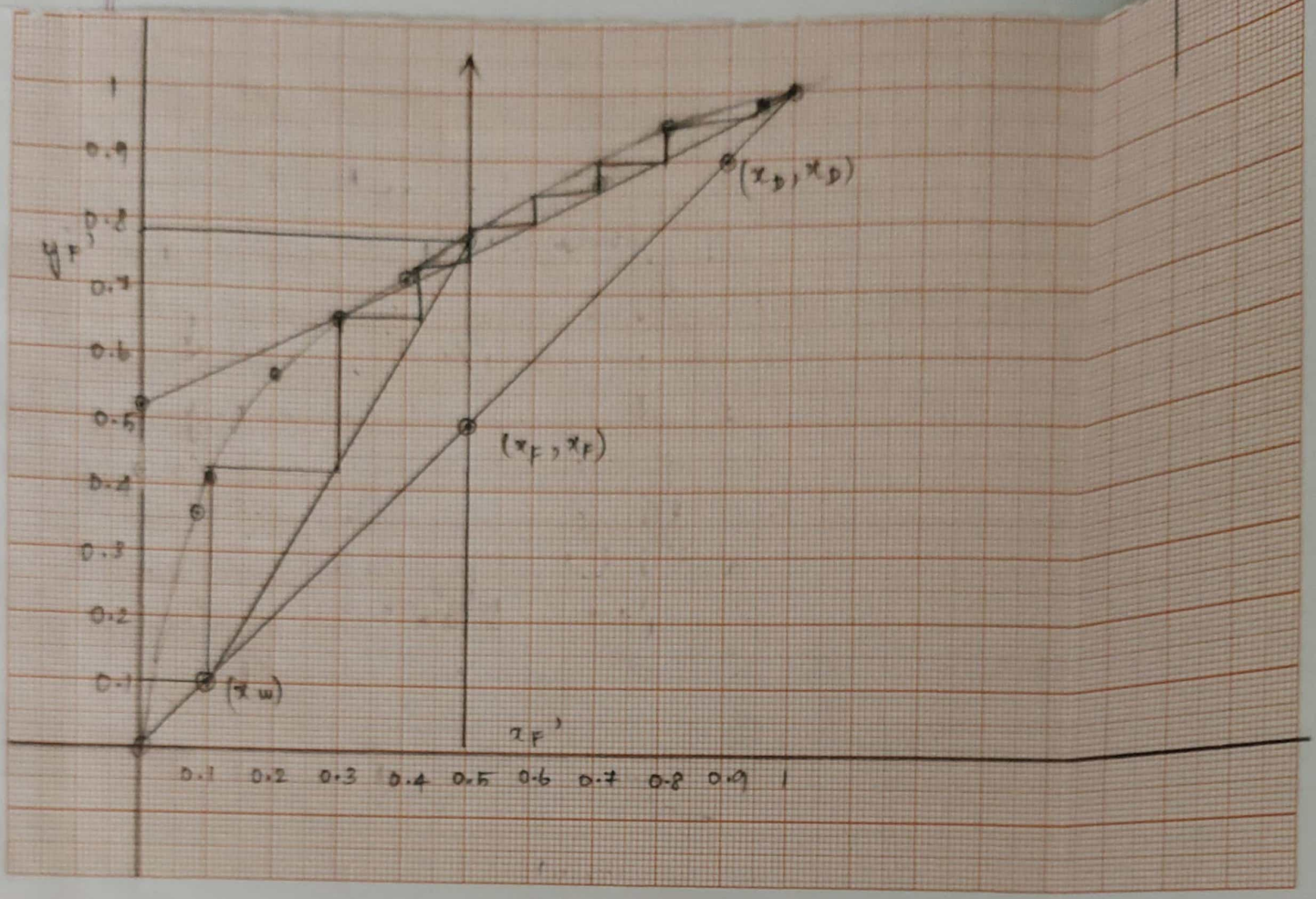
Total no. of stages (n) = 7
(including reboiler)

No. of stages = $n - 1 = 7 - 1 = 6$

Actual stages = 12, efficiency 75%.

No. of stages = $12 \times 0.75 = 9$

The bubble cap fractionating column not removing 100% solute molecules from the aq. methanol for the given recovery over head product, the column available is satisfactory.



$$\frac{R}{R+1} = \frac{0.714}{[0.714+1]}$$

$$= 0.52.$$

Total no. of. stages (n) = 7
(including reboiler)

No. of. stages = n-1 = 7-1 = 6

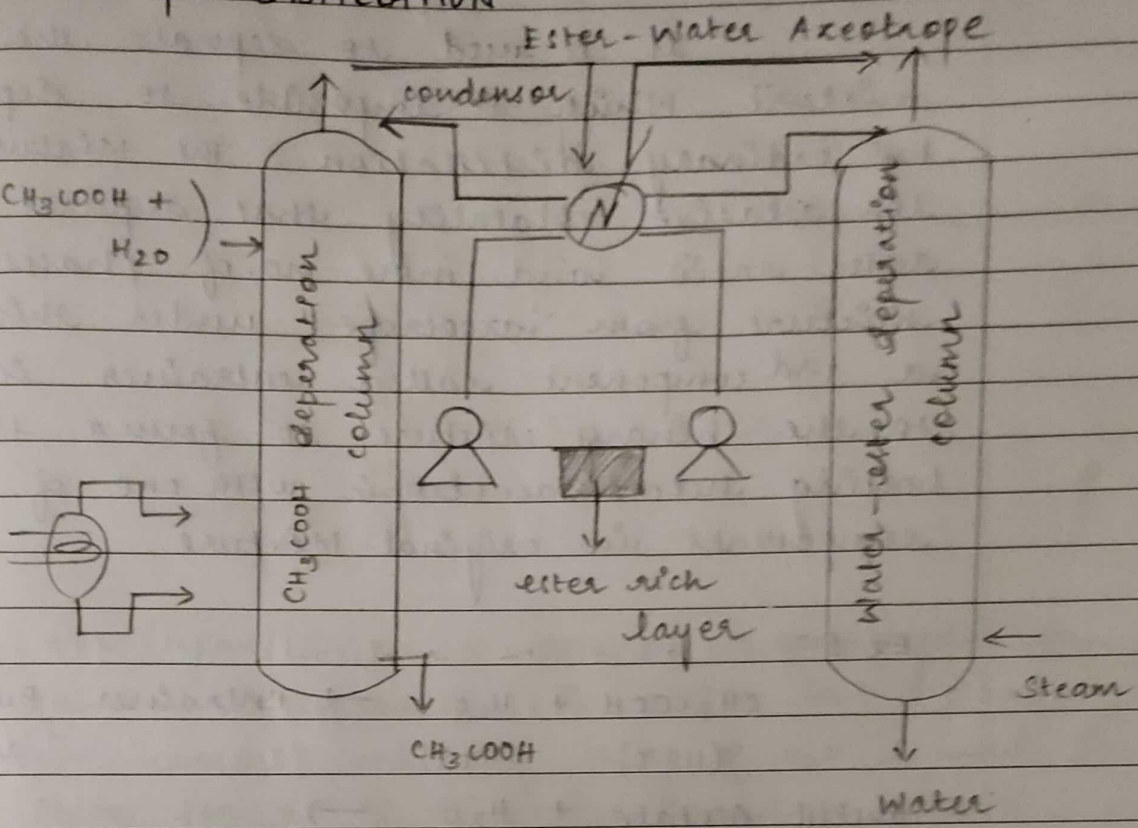
Actual stages = 12, efficiency 75 %.

No. of. stages = 12 x 0.75 = 9.

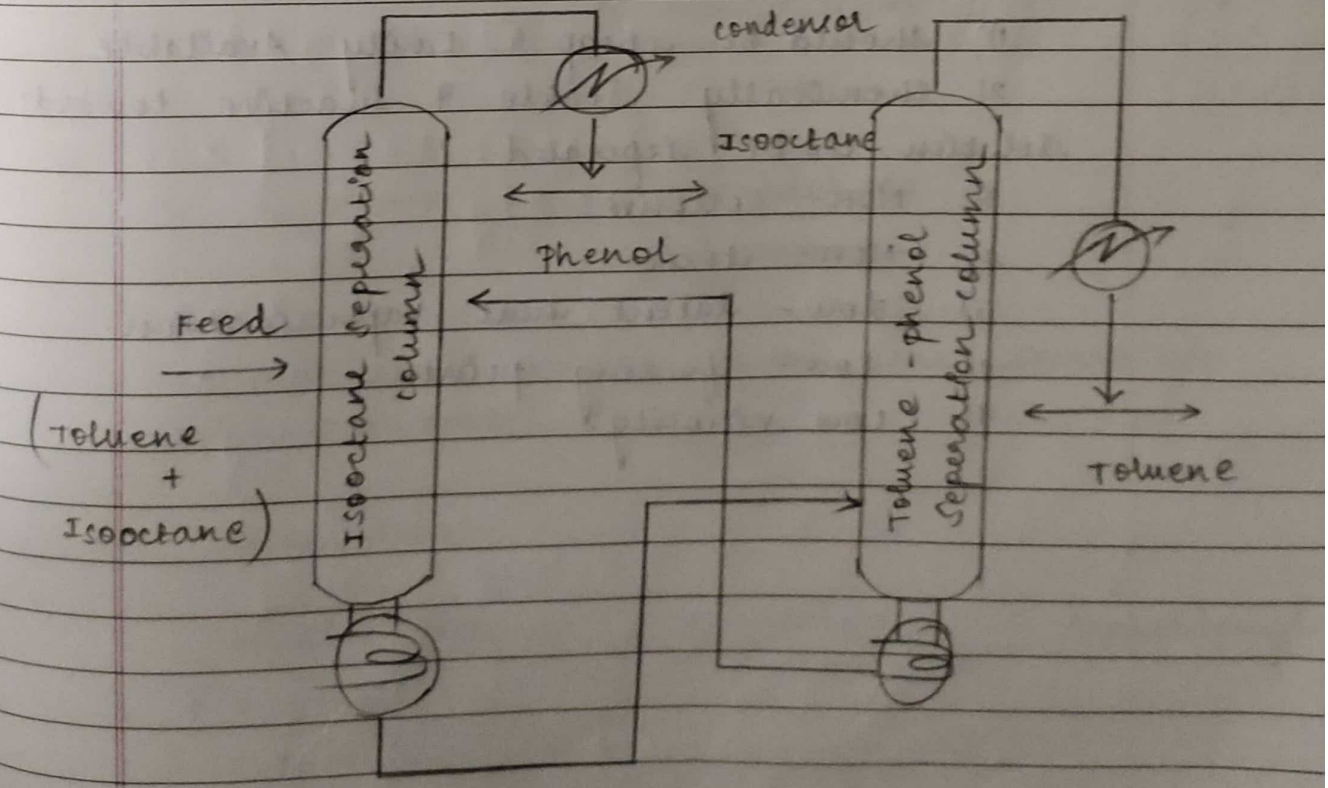
The bubble cap fractionating column not removing 100 % solute molecules from the aq. methanol for the given recovery over head product, the column available is satisfactory.

24/10/22

AZEOTROPIC DISTILLATION :



Extraction Distillation :



Azeotropic Distillation:

It is used to seperate the binary mixture which is impossible to seperate by ordinary distillation. The Mixture with low relative volatality that requires high reflux ratio and more no. of. phases & the mixture from azeotrope. under this condition a 3rd component called entrainer is added to the binary Mixture to form a new low boiling hetero azeotropic with one of the components in original Mixture.

Ex 1:

CH_3COOH + H_2O → entrainer Butyl acetate

(118°C) (100°C)

Butyl Acetate + H_2O →

Relative Volatality ↑, Boiling point ↓.

Properties of entrainer:

1) Should be cheap & Easily Available.
2) Chemically Stable & Inactive towards Solution to be seperated.
3) Non- corrosive.
4) Non- toxic
5) Low- latent heat vapourization
6) Low- freezing point
7) Low viscosity.

UNIT - IV

Extraction operations:

propanol

Liquid - Liquid Extraction:

1. LLE also known as solvent extraction (or) partitioning & is a method to seperate compounds (or) metal complexes based on their relative solubilities in 2 immiscible liquid usually water (polar) & an organic solvent (non-polar).

2. There is a net transfer of one (or) more species from one liq to another liq phase (aq. to organic).

3. The transfer is driven chemical potential, i.e., once the transfer is complete, the overall system of protons & electrons that make up the solutes & solvents are in a more stable configuration (lower free energy).

4. The solvent that is enriched in solutes is called extract.

5. The feed solution that is depleted in solutes is called raffinate

Uses:

Eg 1:

Hydrometallurgical perspective → uranium, plutonium, zirconium & hafnium, cobalt, Nickel.

(2) LLE → production of fine organic

or processing of perfumes, production for biodiseal & vegetable oil.

Distribution coefficient : (Kd)
(also partition coefficient)

→ Kd is equal to the conc. of solute in the organic phase/ divided by its conc. in its aqueous phase.

$$Kd = \frac{\text{conc. of. solute in op}}{\text{conc. of. aq.p}}$$

It depends on Temp. & conc.

Seperation factor : (SF)

It is a measure of the ability of the syst. to seperate 2 solutes.

$$SF = \frac{D_A}{D_B}$$

where,

D_A → distribution ratio of compo

D_B → ,,

Decontamination Factor:

It is used to express the ability a process to remove a contaminant fro a product.

plait point : (P)

The (P) is the intersection of raffinate phase & xtract phase bound curve

10M

Selection of solvent for extraction:

1. Selectivity (β):

The effectiveness of solvent B for seperating a soln of A & B into the components is measured by comparing the ratio of C to A in B-rich phase to that in the A-rich phase at equilibrium. β is analogous to the relative volatility of distillation. If E & R the equilibrium phase.

$$\beta = \frac{(\text{Wt fraction C in E}) / (\text{Wt fraction A in E})}{(\text{Wt fraction C in R}) / (\text{Wt fraction A in R})}$$

$$\beta = \frac{y_E^* (\text{Wt fraction A in R})}{x_R (\text{Wt fraction A in E})}$$

For all usefull extraction the selectivity must exceed unity, the more so the better. If the selectivity is unity, no seperation is possible.

2. Distribution coefficient: (Dc)

The D.C be larger than 1, large values are very desirable since less solvent will then be required for extraction.

③ Insolubility of solvent:

The solvent which is the more insoluble of 2, will be the more useful, The solute is less insoluble in solvent, the capacity of solvent, ~~the capacity of~~ to extract solute in small & large amount of solvent are then required.

④ Recoverability:

→ solvent recovered & recuse & done by MT operations (distillation).

→ If distillation is to be used, the solvent should show high relative volatality for low-cost recovery.

→ That substance in the extract, either solvent or solute, which is present as the less or quantity should be more volatile in order to reduce heat costs.

→ If solvent must be volatilized, its latent heat or vapourization should be small. 13/10/22

⑤ Density:

→ A diff. in densities of the saturated liq. phases is necessary, both for stage wise & continuous-contact equipment operation.

→ The density diff. for equilibrium phase will become less as solute conc. increases & will be zero at plait point.

⑥ Interfacial tension:

It should be high & it falls to zero at the plait point.

⑦ Chemical ~~reacti~~ relativity:

The solvent should be stable chemically & in toward the other components of the system & construction material.

⑧ Viscosity, Vapour pressure & Freezing point:

These should be low for ease in handling & storage.

⑨. The solvent should be non-toxic, Non-flammable & of low cost.

13/10/22

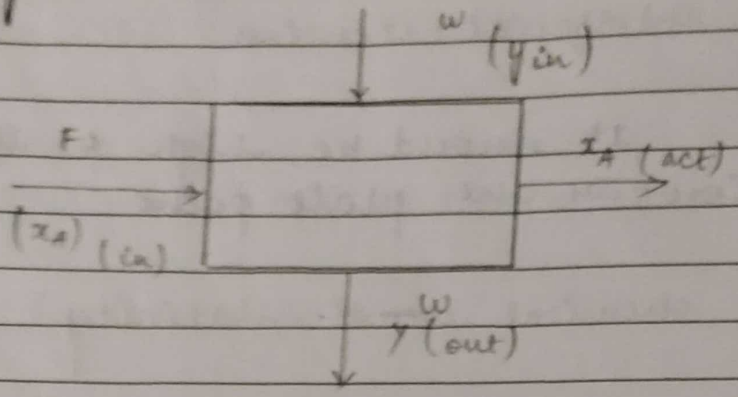
Simple (or) Single Stage (or) Batch extraction

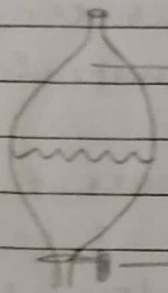
In batch extraction solution containing a desired solute is mixed with the solvent invisible with the soln. In the common phase, Initial soln is dil. and aqueous and immisible solvent and organic.

The simple extraction are the batch

extraction, they are analysed by graphical and analytical method.

Analytical Method:





Solute conc. in the lighter phase x to the solute conc. in the heavier phase.

$$x \propto y$$

$$x = ky \longrightarrow (1)$$

$k \rightarrow$ distribution coeff (or) equilibrium constant.

$x \rightarrow$ Solute conc. in the lighter phase

$y \rightarrow$ Solute conc. in the heavy phase

Mass balance on solute,

Feed = output.

$$H y_F + L x_{(=0)} = Hy + Lx$$

$$H y_F = Hy + Lx \longrightarrow (2)$$

$$x = k.y \longrightarrow (3)$$

Sub (1) in (2)

$$Hy_F = Hy + LKy$$

$$Hy_F = y\,(H + Lx)$$

$$y = \frac{Hy_F}{(H + Lk)}$$

14/10/22

$$y = \left[\frac{H}{H + LK}\right] y_F$$

RHS : x & ÷ by H.

$$y \left[\frac{H/H}{[H/H] + [LK/H]}\right] y_F$$

$$y = \left[\frac{1}{1 + LK/H}\right] y_F \longrightarrow (4)$$

$$\because E = \frac{LK}{H}$$

E - Extraction factor.

Sub value of 'E' in equn (4)

$$y = \left[\frac{1}{1 + E}\right] y_F$$

$$y = \frac{y_F}{1 + E} \longrightarrow (5)$$

To find x:

$$H y_F = H.y + L.x$$

$$y = \frac{x}{k}$$

$$x = k.y$$

$$H y_F = H \frac{x}{k} + L.x$$

$$H y_F = \left[\frac{H}{k} + L\right] x$$

$$x = \frac{H y_F}{\left[\frac{H}{k} + L\right]}$$

$$x = \frac{H y_F}{\left[\frac{H + LK}{k}\right]}$$

$$x = \frac{H.y_F\, k}{H + LK}$$

$$x = \left[\frac{H}{H + LK}\right] y_F.k$$

TO RHS $\times$ & $\div$ by 'H'

$$x = \left[\frac{H/H}{(H/H) + \left(\frac{LK}{H}\right)}\right] y_F.k$$

$$x = \left[\frac{1}{1 + LR/H}\right] y_F \cdot K$$

$$E = \frac{LR}{H}$$

$$x = \left[\frac{1}{1+E}\right] y_F \cdot K$$

$$x = \frac{y_F \cdot K}{1+E} \longrightarrow (6)$$

K is large, most of the solute wines up in extraction solvent.

Fraction extracted (p)

$$p = \frac{L \cdot x}{H \cdot y_F} = \frac{E}{1+E} \longrightarrow (7)$$

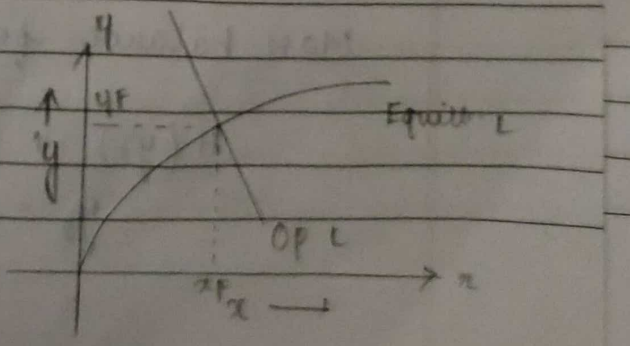
2) Graphical Method :-

For equilibrium line

$$x = x(y) \longrightarrow (8)$$

For operating line.

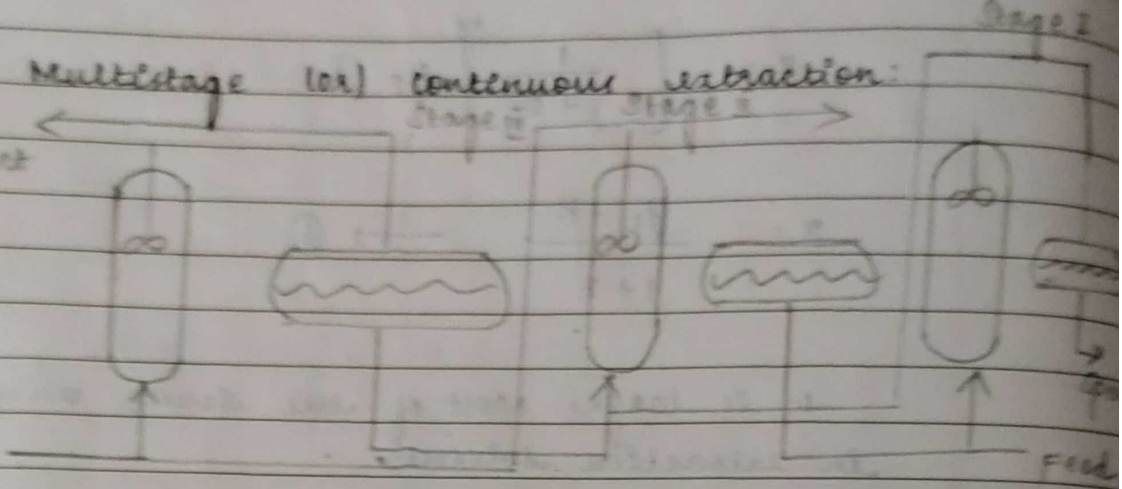
$$x = \left(\frac{H}{L}\right)(y_F - y) \longrightarrow (9)$$



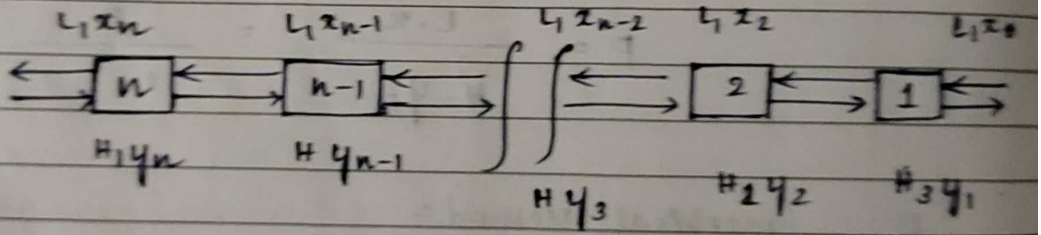
From equⁿ ② & ①

Can be plotted a coordinate / there intersection gives the values of x & y after extraction.

Multistage (or) continuous extraction:



Countercurrent operation:



Analytical Method:

Equilibrium R/s is $x_n = k y_n$ → ①

This assumes that the solute conc. in the light & heavier liq. are in the equilibrium in each stages.

Mass balance for 1st stage:

$$H(y_2) + L(0) = Hy_1 + Lx_1 \rightarrow ②$$

$$y_2 = (E+1)\, y_1 \rightarrow ③$$

For 2nd stage,

$$H y_3 + L x_1 = H y_2 + L x_2 \longrightarrow (4)$$

$$y_3 = (1+E)\, y_2 - E$$

$$y_3 = (1+E)(E+1)\, y_1 - E$$

$$y_3 = (1+E+E^2)\, y_1 \longrightarrow (5)$$

For n^{th} stage,

$$y_{n+1} = (1+E+E^2+\dots+E^n)\, y_1 \longrightarrow (6)$$

$$y_{n+1} = \frac{E^{n+1}-1}{E-1} \longrightarrow (7)$$

The equn (7) that relates the feed conc. y_{n+1} to the exhausted effluent conc. y_1 & the relation is based on no. of. stages (n) & Extraction factor (E)

Fraction extracted (P),

$$P = \frac{L \cdot x_n}{H\, y_{n+1}}$$

$$P = \frac{1 - y_1}{y_{n+1}}$$

$$P = \frac{E(E^n - 1)}{E^{n+1} - 1} \longrightarrow (8)$$

If $E = 1$, then P is $n/n+1$

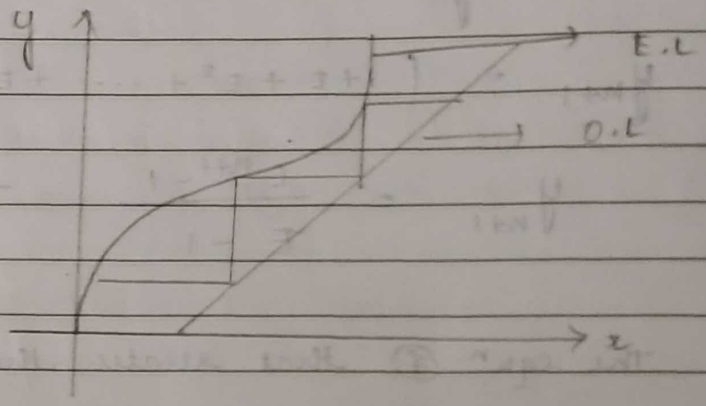
Graphical Method:

Eqn for Equilibrium line

$$x_n = x_n(y_n) \longrightarrow (9)$$

Eqn for operating line

$$x_n = \frac{H}{L}(y_{n+1} - y_1) \longrightarrow (10)$$



15/10/22 Solid Liquid Extraction (Leaching)

Leaching:

It is seperation of solute from solid mixture by dissolving it in liquid phase.

Steps:

1) Solute ⟶ Solvent (leaching)
2) Seperation of Solute in Solvent (washing).

Step-1) contracting Solvent & Solute that affect in transfer of Solute. (Leaching).

2) The seperation of solution from the remaining solid (Washing)

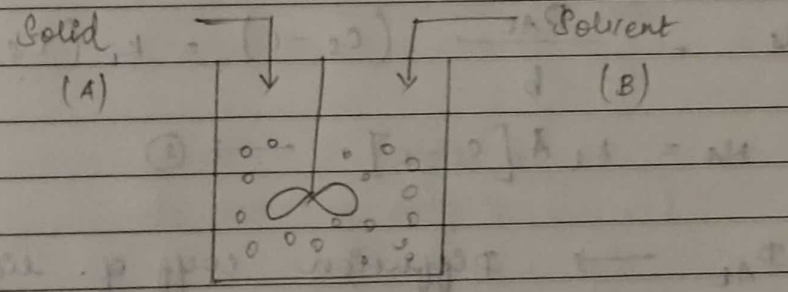
Factors affecting :

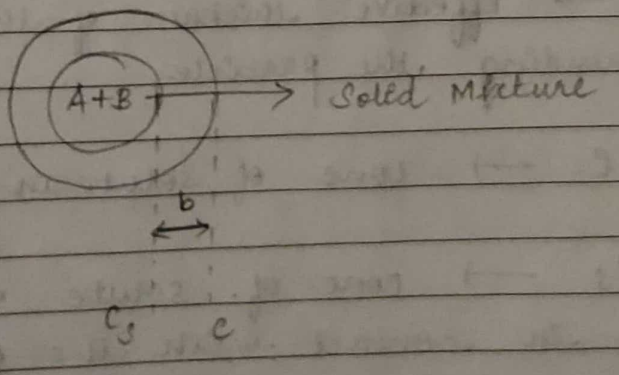
1) particle size
2) Solvent
3) Temperature
4) Agitation.

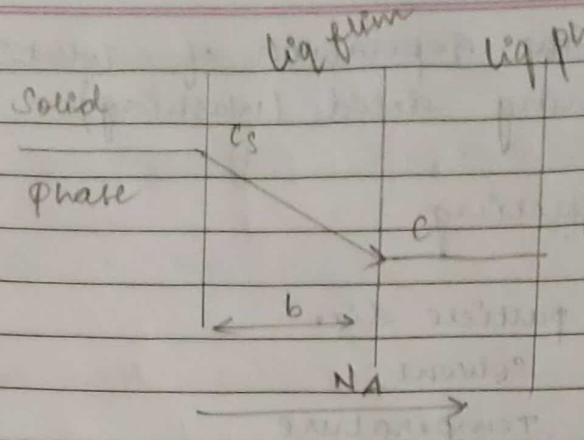
Leaching :

- Batch (or) single staged
- Continuous (or) Multistage
  - co-current
  - counter current.

1) Batch Leaching :







The MT rate within the code is resedue q difficult to resist cox it is impossible to define the ~~simple~~ shape of the channel through which transfer must be taken place.

It is possible to obtain an approximate indication of the rate of transfer of solute from the solid particle to the bulk liq.

i) Leaching:

$$N_A = \frac{D_{AL}}{b}(C_s - C) = k_L(C_s - C) \longrightarrow (1)$$

$$N_A = k_L A[C_s - C] \longrightarrow (2)$$

D_{AL} ⟶ Diffusion coff of. liq. phase

b ⟶ Effective thickness of the liq. film surrounding the particle.

C ⟶ conc. of solute in the bulk liq.

C_s ⟶ conc. of. solute in the saturated soln in contact with the particle.

$$N_A = \frac{dM}{dt} \longrightarrow (3)$$

equate ② & ③

$$\frac{dM}{dt} = K_L A [C_s - C] \longrightarrow ④$$

$$M = CV \longrightarrow ⑤$$

For constant volumes $\Rightarrow dM = V dC \longrightarrow ⑤$

From equⁿ ④

$$\frac{dm}{dt} = K_L A (C_s - C)$$

$$dM = K_L A [C_s - C] dt \longrightarrow ⑥$$

Sub equⁿ ⑥ in equⁿ ⑤

$$K_L A [C_s - C] \cdot dt = V dC$$

$$\frac{dC}{(C_s - C)} = \left[\frac{K_L \cdot A}{V}\right] dt$$

$$C \longrightarrow 0 \longrightarrow C_s$$

$$t \longrightarrow 0 \longrightarrow t$$

$$\int_0^{C_s} \frac{dC}{(C_s - C)} = \int_0^t \left[\frac{K_L \cdot A}{V}\right] dt$$

$$\ln\left[\frac{C_s - C}{C_s}\right] = \frac{-K_L \cdot A}{V} t$$

$$\frac{C_s - C}{C_s} = e^{\left[\frac{-K_L A}{V} t\right]}$$

$$C_s - C = e^{\frac{-K_L A t}{V}} C_s$$

$$C = C_s + C_s\, e^{\frac{-K_L A t}{V}}$$

$$C = C_s \left[1 + \left[e^{\frac{-K_L A t}{V}}\right]\right] \longrightarrow (7)$$

ii) Washing:

$$\text{No. of Washing } (\theta_N) = \frac{1}{(1+\phi)^N} \longrightarrow (8)$$

$$\phi = b/a \longrightarrow (9)$$

a → Solvent remaining

θ_N → Fraction of solute remains with the residue from the original.

ϕ → Solvent decanted per solvent remaining in the insoluble solvent.

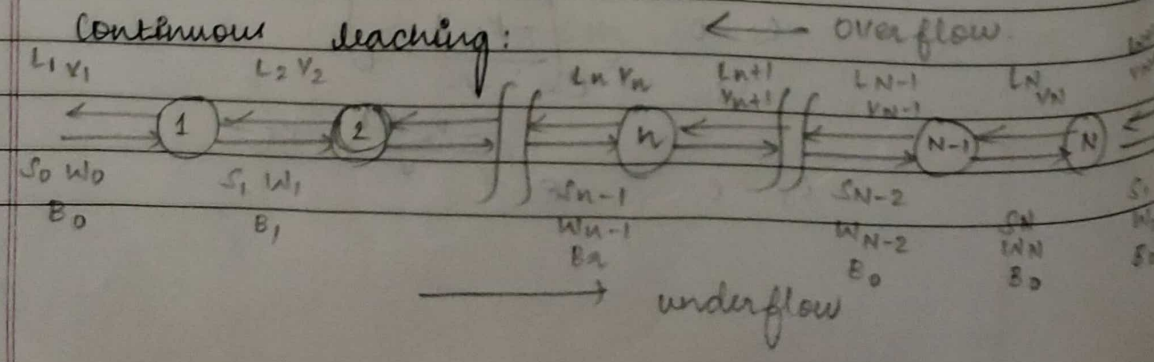
b → Solvent Decanted.

$$\theta_N = \frac{S_N}{S} \longrightarrow (10)$$

S_N → Wt. of solute remaining in the solute after washing.

S → Original wt of solute which was in the solid before washing.

Continuous leaching:



L → Is the flow solute in overflow.

S → Is the flow of solute in underflow

V → Is the flow solution in overflow

W → " " in under flow.

B → Is the flow insoluble solid in underflow

Case - I

Constant UF.

The amt of solvent removed with the insoluble solid in the underflow is constant & independent of the conc. of soln in the thickner. Then the amt of solvent leaving each thickner in the underflow will be the same and also the amt of solvent in overflow will be flow. Hence the ratio of the solvent discharge in the overflow to the underflow is constant.

Solvent Discharge:

$$\text{Recovery } (R) = \frac{\text{Amt of solvent Discharge in overflow}}{UF}$$

$$= \frac{V-L}{W-S}$$

$$R = \frac{\text{Amt of. Solute Discharge in UF}}{UF}$$

$$= L/S$$

Solution Discharge:

$$R = \frac{V}{W}$$

Material Balance on stage 'n'

$$(S_{n-1}) + (L_{n+1}) = L_n + S_n \longrightarrow (1)$$

Where, $L = RS$

$$L_n = RS_n \longrightarrow (2)$$

$$L_{n+1} = RS_{n+1} \longrightarrow (3)$$

$$(S_{n-1}) + R(S_{n+1}) = [RS_n] + S_n \longrightarrow (4)$$

$$\frac{1}{R} S_{n-1} + S_{n+1} = S_n + \frac{1}{R} S_n$$

$$\frac{1}{R}(S_{n-1}) + (S_{n+1}) = S_n\left(1 + \frac{1}{R}\right)$$

$$(S_{n+1}) - \left[S_n\left(1 + \frac{1}{R}\right)\right] + \frac{1}{R}(S_{n-1}) = 0$$

$$S_{n+1} = ES_n$$

$$S_{n-1} = E^{-1}S_n$$

$$ES_n - \left[1 + \frac{1}{R}\right] S_n + \frac{1}{R}(E^{-1}S_n) = 0 \longrightarrow (5)$$

equn (5) $\times E^1$

$$E^2 S_n - \left[1 + \frac{1}{R}\right] S_n E + \frac{1}{R} S_n = 0$$

Change $E \longrightarrow \rho$

$$\rho^2 - \left(1 + \frac{1}{\rho}\right)\rho + \frac{1}{R} = 0$$

2nd order equⁿ:

$$S_n = A\rho_1^n + B\rho_2^n \longrightarrow (6)$$

$$\rho_1 = 1 \qquad \rho_2 = 1/R$$

$$S_n = A + B\left(1/R\right)^n \longrightarrow (7)$$

using Boundary condition I:

Where, $n=0$

$$S_n = S_0$$

$$S_0 = A+B\left(\frac{1}{R}\right)^0$$

$$S_0 = A+B \longrightarrow (8)$$

B.C : (2) :

Where $n=1$, $S_n = S_1$

$$S_1 = A+B\left(1/R\right)^1 \Rightarrow S_1 = A+B\left(1/R\right)$$

$$L = RS$$

$$L_1 = RS_1 \Rightarrow S_1 = L_1/R$$

$$L_1/R = A+B\left(1/R\right)$$

$$L_1/R = R\left(RA+B\right)$$

$$L_1 = RA+B \longrightarrow (9)$$

Subtraction of equⁿ (9) - equⁿ (8)

$$\left[L_1 - S_0\right] = \left(RA+B\right) - \left(A+B\right)$$

$$\left(L_1 - S_0\right) = A\left(R-1\right) \longrightarrow (10)$$

$$A = \frac{L_1 - S_0}{R-1} \longrightarrow (11)$$

If you are using pure solvent

$$L_{n+1} = 0$$

$$L_1 = S_0 - S_n \longrightarrow (12)$$

Sub (12) in (11)

$$A = \frac{S_0 - S_n - S_0}{R-1} \qquad A = \frac{-S_n}{R-1} \longrightarrow (13)$$

$$S_0 = A + B \qquad S_0 = \left[\frac{-S_n}{R-1}\right] + B$$

$$B = S_0 - \left[\frac{-S_n}{R-1}\right]$$

$$B = S_0 - \left[\frac{L_1 - S_0}{R-1}\right] \Rightarrow B = \frac{S_0(R-1) - L_1 + S_0}{(R-1)}$$

$$B = \frac{S_0(R-1) + S_n}{R-1} \longrightarrow (14)$$

$$S_n = A + B\left(\frac{1}{R}\right)^n$$

$$S_n = \frac{AR^n + B}{R^n}$$

$$S_n R^n = AR^n + R \longrightarrow (15)$$

Sub (A) & (B) in equn (15)

$$\frac{S_n}{S_0} = \frac{(R-1)}{(R^{n+1}-1)} \longrightarrow (16)$$

Fractional solute discharge $(f) = \frac{S_n}{S_0}$

$$f = \frac{S_n}{S_0} = \frac{(R-1)}{(R^{N+1}-1)}$$

Recovery $= 1 - f$

$$N = \frac{\log\left[1 + \left(\frac{R-1}{f}\right)\right]}{\log R}$$

Case - II

Variable under Flow

(i) → Diagram:

$W_1 \neq W_2 \neq W_3 \neq W_n \neq W_N$ → soln u.F variable

$$X = \frac{L}{V} = \frac{S}{W} = \frac{\text{kg solute}}{\text{kg soln}} = \text{Wt. fraction}$$

M.B on 'N' section:

$$S_{n-1} + L_{N+n} = L_n + S_N \longrightarrow (1)$$

Soln MB on 'N' section

$$W_{n-1} + V_{N+1} = V_A + W_N \longrightarrow (2)$$

from equn (1)

$$L_n = L_{N+1} + S_{n-1} - S_W \longrightarrow (3)$$

From equn (2)

$$V_n = V_{N+1} + W_{n-1} - W_N \longrightarrow (4)$$

UNIT - 4

1) 5 % (by wt), 5 % of soln in acetaldehyde in toluene is extracted in water in 3 stage cross current unit, if 100 kg of H_2O is used per stage for 500 kg of feed. Calculate the % extraction of acetaldehyde & the wt. of final raffinate & extract the equilibrium eq is given as $y = 2.3x$

$y \rightarrow$ $\frac{\text{kg of acetaldehyde}}{\text{kg of water}}$

$x \rightarrow$ $\frac{\text{kg of acetaldehyde}}{\text{kg of toluene}}$

Assume that toluene & H_2O are immiscible with each other.

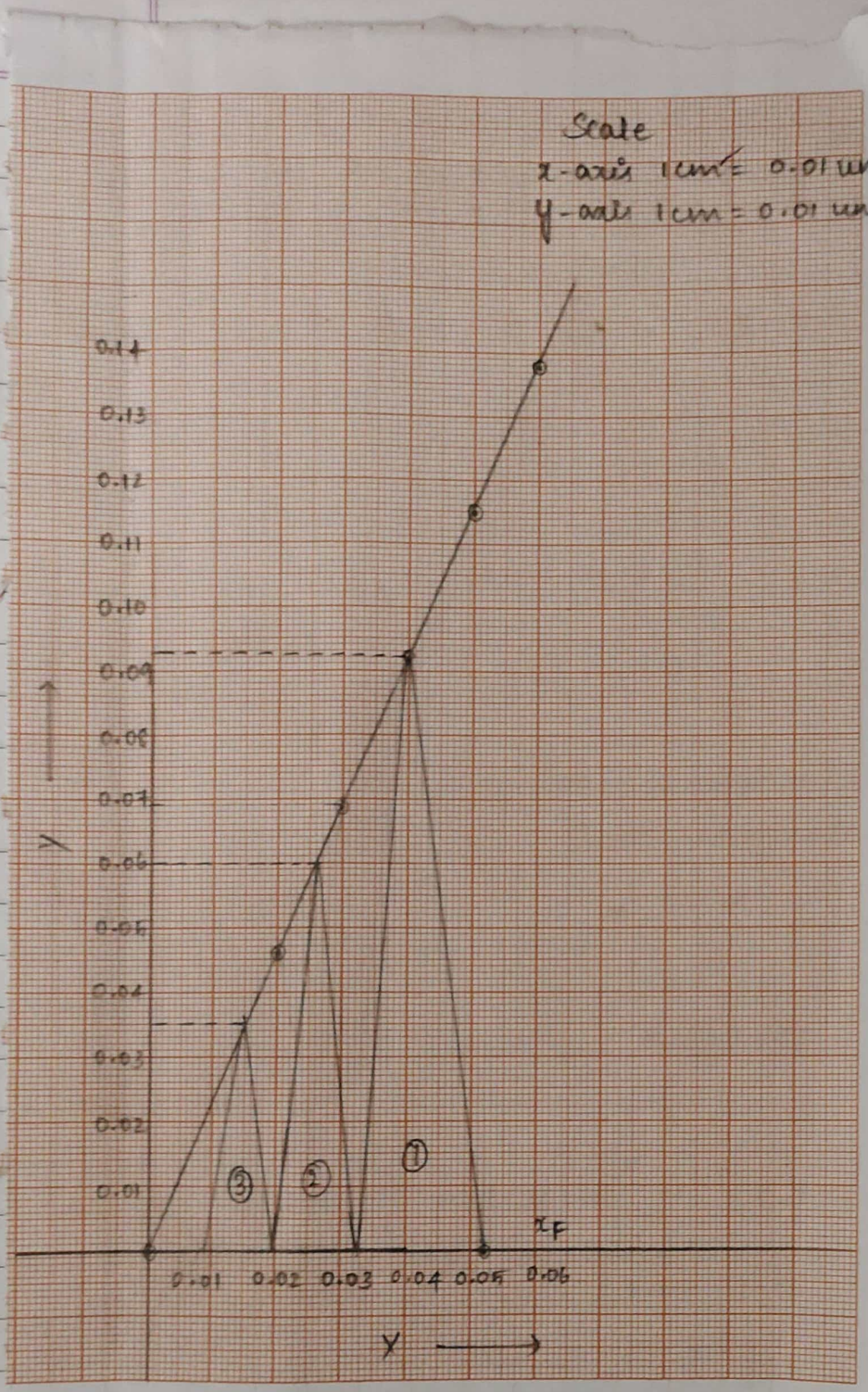
Given data:

A → Toluene, B - Water; C - Acetaldehyde

$F = 500$ kg

$x_F = 5\%$ (by wt)

$\therefore x_F = 0.05$



0.05
0.01
0.04

$x_F = 0.05$

$x_F = \frac{x_F}{1 - x_F}$

$= \frac{0.05}{1 - 0.05}$

$= 0.0526$

475 kg

$= 475 \times 0.019$

$= 9.025$ kg

$(475 + 9.025)$ kg

484.025 kg.

$- y_3) \times 100$

$2] \times 100$

Amt of c in final stage $= y_3 \times 100$

$= 0.02 \times 100$

$= 2$ kg

Total wt of extract $= 300$ kg $+ 17.9$ kg

$= 317.9$ kg

% Extraction $= \left(\frac{17.9}{25}\right) \times 100$

0.05
0.01
0.06

B = 100 kg / each stage

Eq. relationship $y = 2.3x$

$X \longrightarrow 0 - 0.06$

$y = 2.3x$ $\quad x = 0$

$y_1 = 2.3(0) = 0$

$y_2 = 2.3(0.01) = 0.023$

$y_3 = 2.3(0.02) = 0.046$

$y_4 = 2.3(0.03) = 0.069$

$y_5 = 2.3(0.04) = 0.092$

$y_6 = 2.3(0.05) = 0.115$

$y_7 = 2.3(0.06) = 0.138$

$x_F = 0.05$

$x_F = \frac{x_F}{1 - x_F}$

$= \frac{0.05}{1 - 0.05}$

$= 0.0526$

The Wt. of the raffinate is 475 kg

Final raffinate (x_3) $= 475 \times 0.019$

$= 9.025$ kg

Total Wt. of raffinate $= (475 + 9.025)$ kg

$= 484.025$ kg.

Total 'c' extracted $= (y_1 + y_2 + y_3) \times 100$

$= [0.09 + 0.06 + 0.02] \times 100$

$= 17.9$ kg

Amt of 'c' in final stage $= y_3 \times 100$

$= 0.02 \times 100$

$= 2$ kg

Total Wt. of extract $= 300$ kg $+ 17.9$ kg

$= 317.9$ kg

% Extraction $= \left(\frac{17.9}{25}\right) \times 100$

% Extraction = 71.6 %.

2) The H_2O dioxane soln is to be seperated by extraction process using benzene as solvent at 25 °C, the equ. distribution of dioxane of H_2O & benzene is as follows:

Wt % of Dioxane in H_2O	5.1	18.9	25.2
Wt % of Dioxane in benzene	5.2	22.5	32

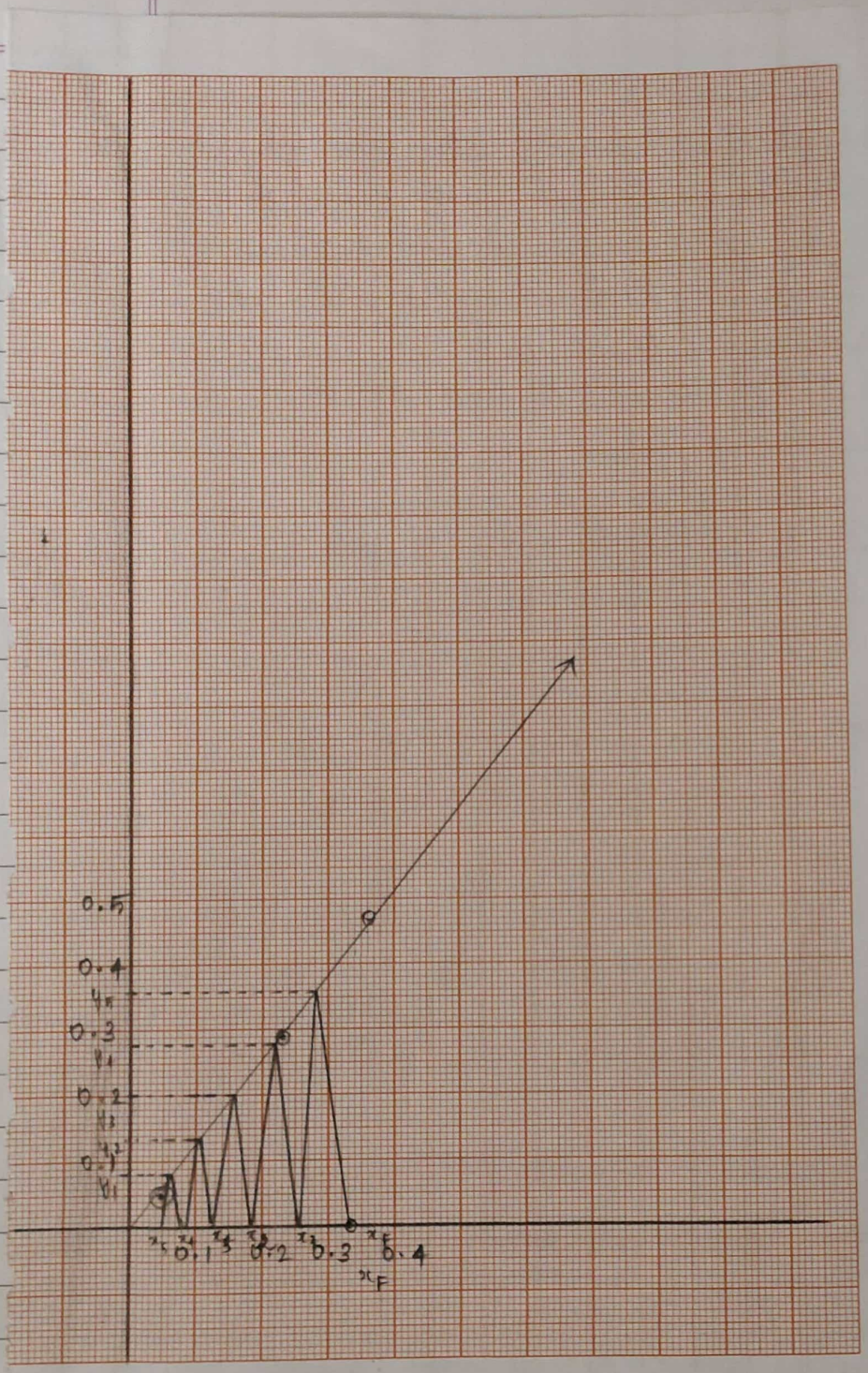
At there conc. H_2O & benzene are substantially insoluble.

1000 kg of 25 % dioxane H_2O soln, is to be extracted to be remove 95 % of dioxane the benzene used for this process is dioxane free.

i) calculate the benzene requirement for a single batch operation.

ii) calculate the benzene req. for the 5 stage cross current of 600kg solvent used in each stage:

Wt % of dioxane H_2O	Wt % of Dioxane benzene	x	y	$X = \frac{x}{1-x}$	$Y = \frac{y}{1-y}$
5.1	5.2	0.051	0.052	0.054	0.054
18.9	22.5	0.189	0.225	0.233	0.290
25.2	32	0.252	0.32	0.336	0.471



C = 250kg

ane

$\frac{0.25}{-0.25} = 0.33$

65

graph

= 0.4

= 0

= 0.33

= 0.0165

B = 956.25 kg

$$\% \text{ recovery} = \frac{(0.33 - 0.05)}{0.33} \times 100$$

% recovery =

i) F = 1000 kg A = 750 kg C = 250 kg

B = ?

A = H_2O, B = C_6H_6, C = Dioxane

X_F = 25 % = 0.25 $x_F = \frac{0.25}{1-0.25} = 0.33$

$X_{RNP} = 0.05\ (0.33) = 0.0165$

$Y_{in} = 0$

$A/B = \frac{Y_1 - Y_S}{X_F - X_{NP}}$

From graph

$Y_1 = 0.4$

$Y_S = 0$

$X_F = 0.33$

$X_{NP} = 0.0165$

$\frac{750\ kg}{B} = \frac{(0.4 - 0)}{(0.33 - 0.0165)}$ $\Rightarrow$ B = 956.25 kg

ii) 600 kg of solvent

$A/B = \frac{750}{600} = 1.25$

Final conc. of solute (x_F) = 0.05

% recovery $= \frac{(0.33 - 0.05)}{0.33} \times 100$

% recovery =

375 Pg No:

3) Nicotine in a H_2O soln containing 1% Nicotine is to be extracted, once with kerosene @ 20°C. Kerosene & H_2O are insoluble. Determine the % extraction, if 1000 kg of feed soln is extracted once with 1500 kg solvent. What will be the extraction if 3 ideal stages are used with 500 kg solvent in each stage

x	0	0.00101	0.00246	0.00502	0.00751	0.00998	0.0204
y	0	0.00081	0.001962	0.00456	0.00686	0.00913	0.0187

Where

x → kg Nicotine / kg water

y → kg Nicotine / kg kerosene.

4) calculate Min. solvent required in kg/h if the extract is done in counter current fashion. Estimate the no. of stages needed if 1.5 times the min. amt of solvent is used.

5) 1000 kg/h of a water dioxane soln containing 20% dioxane to be continuously and counter current Extracted with C_6H_6 at 25°C to recover 80% dioxane H_2O and benzene are essentially insoluble and the equi. distract of dioxane b/w.

Wt % Dioxane in H_2O	5.1	18.9	25.2
" C_2H_6	5.2	22.5	32

Determine the No. of. theoratical stages if the solvent rate is 1.5 times the Min.

6) 500 kg of the inert solid containing 28% by mass of the H_2O soluble component (A) is agitated with $100 m^3$ of water for 600 sec. After each decanting 25% of the soln remains in the residue water is saturated with the solute at a conc. of $2.5 kg/m^3$. Find the conc. of the solute A in the soln after the leaching and no. of Washing such that the conc. of A in the solid residue 0.01% by mass. In a pilot scale test using a vessel $1 m^3$ in volume a solute was leached from an inert solid & the H_2O was 75% sat. in 10s. Assume eqn. to those in pilot scale vessel.

7) Repeat (6) but the time of leaching is 300s, Find the conc. of solute in the solid (dry basis) after five Washes where the decanting ratio is the same.

3/11/22 UNIT - 5

Problem for

A 5 Normal

1. Oxalic acid in H_2O is adsorbed using 1g of activated carbon at 25 °C. In the following equilibrium condition.

Initial conc.	0.1	0.2	0.3	0.4	0.5
Final conc.	0.002	0.008	0.12	0.21	0.26

using adsorption isotherm find the Freudlich constant K_f & n

$m = 1g \quad T = 25°C$

$\log(x/m) = \log k + n \log ce$

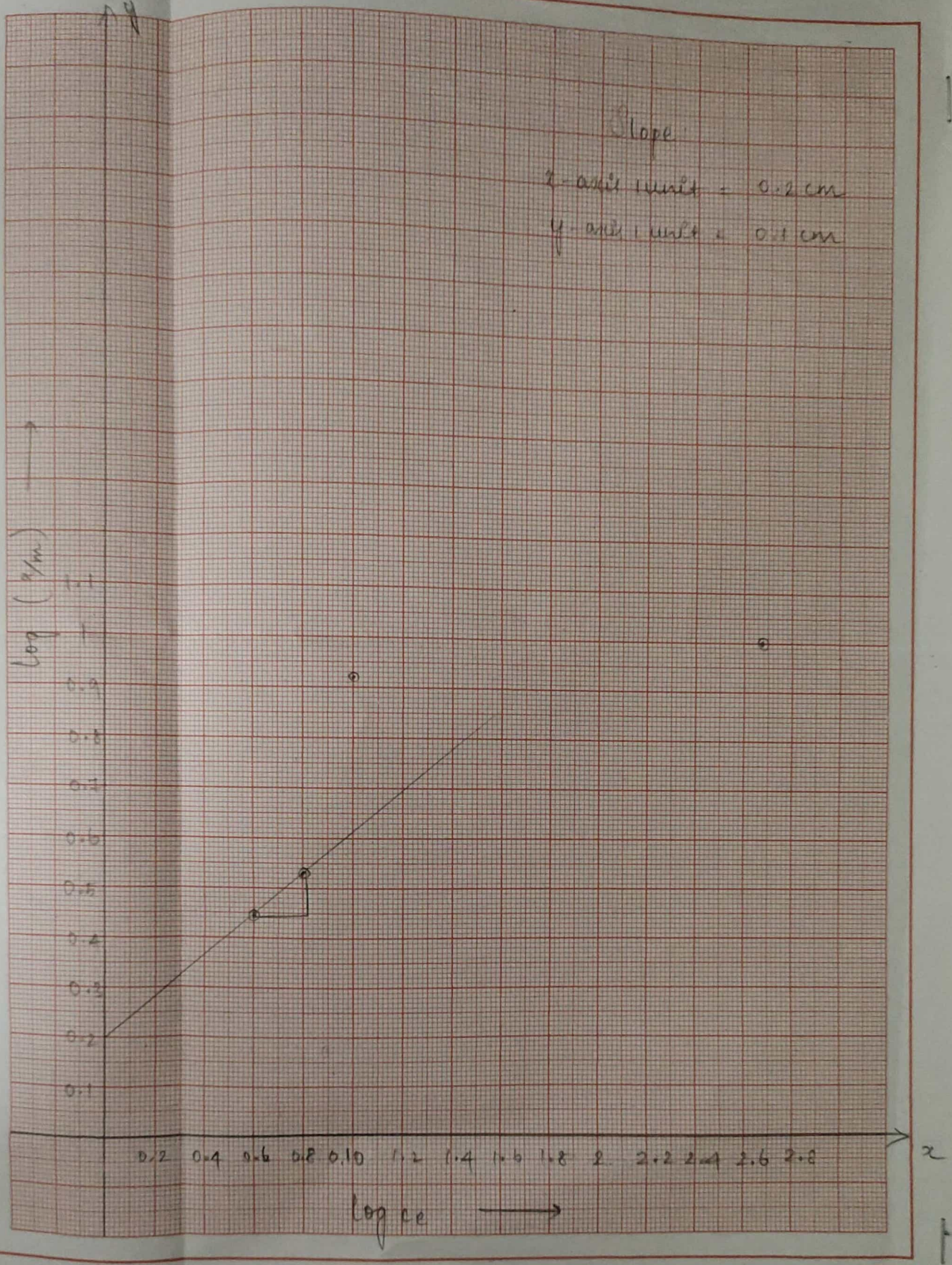
C_0	C_E	$x = (C_0 - C_E)$	m	x/m	$\log x/m$	$\log ce$
0.1	0.002	0.098	1	0.098	1.008	2.696
0.2	0.008	0.192	1	0.192	0.920	1.096
0.3	0.12	0.18	1	0.18	0.744	0.920
0.4	0.21	0.19	1	0.19	0.721	0.677
0.5	0.26	0.24	1	0.24	0.619	0.585

Slope = 0.69

Intercept = 1.659

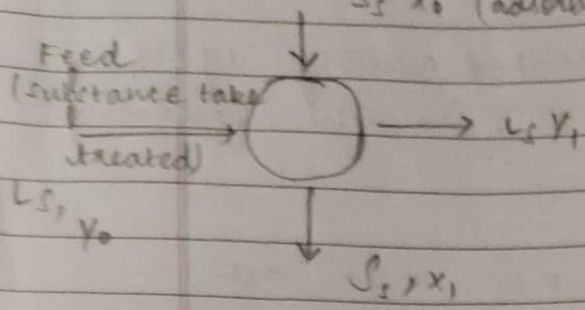
8/11/22

UNIT - 5



Shine

Single stage adsorption



L_s → Mass of the solvent

S_s → Mass of the adsorbent

Y_0 → Adsorbable solute conc. in the soln.

Y_1 → Adsorbable solute conc. in the soln leaving the state

S_0 → Conc. of fresh adsorbent

X_0 → Solute conc. in the adsorbent

X_1 → Solute conc. in the adsorbent leaving the state.

Material balance:

Solute removed from the soln = solute picked up by the adsorbent.

$$L_s(Y_0 - Y_1) = S_s(X_1 - X_0) \longrightarrow (1)$$

$$\frac{(Y_0 - Y_1)}{(X_0 - X_1)} = \frac{-S_s}{L_s} \longrightarrow (2)$$

$$L_s(Y_0 - Y_1) = -S_s(X_0 - X_1)$$

Freundlich isotherm:

$$Y^* = mx^n \longrightarrow (3)$$

At the final equ. stage

$$Y_1^* = mX_1^n$$

$$X_1 = \left[\frac{Y_1}{m}\right]^{1/2} \longrightarrow (4)$$

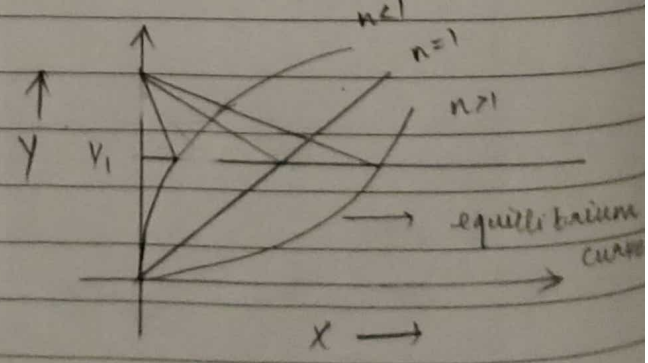
Sub equn (4) in (2)

$$\frac{(Y_0 - Y_1)}{\left(X_0 - \left(\frac{Y_1}{m}\right)^{1/2}\right)} = -\frac{S_s}{L_s}$$

Fresh adsorbent to be dent, so that $X_0 = 0$

$$\frac{(Y_0 - Y_1)}{-\left(\frac{Y_1}{m}\right)^{1/n}} = \frac{-S_s}{L_0} \longrightarrow (5)$$

$$\frac{(Y_0 - Y_1)}{\left(\frac{Y_1}{m}\right)^{1/n}} = \frac{S_s}{L_s} \longrightarrow (6)$$



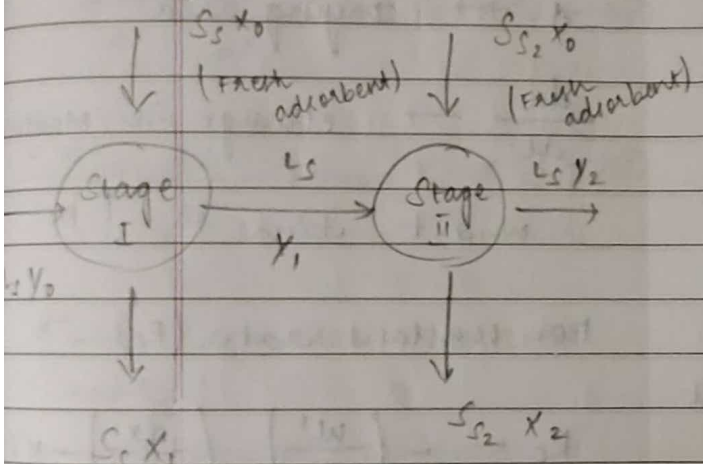
Graph

Based on freundlich isotherm can be applicable.

Note:

1. Generally $n = 2 - 10$ (represents good)
2. $n = 1 - 2$ (Moderately difficult).
3. $n < 1$ (pure adsorption).

Multistage adsorption:



MB for stage - I

$$L_s (Y_0 - Y_1) = S_{s_1} (X_0 - X_1) \rightarrow (1)$$

MB for stage - II

$$L_s (Y_1 - Y_2) = S_{s_2} (X_0 - X_2) \rightarrow (2)$$

Apply freudlich isotherm.

for stage I

$$y^* = x^n m \rightarrow (3)$$

$$X_1 = \left(\frac{Y_1}{m}\right)^{1/n} \rightarrow (4)$$

For stage II

$$X_2 = \left(\frac{Y_2}{m}\right)^{1/n} \rightarrow (5)$$

sub x_1 value in equn (1)

$$L_s (Y_0 - Y_1) = S_{s_1} \left(X_0 - \left(\frac{Y_1}{m}\right)^{1/n}\right)$$

$$X_0 = 0$$

$$\frac{-S_{s_1}}{L_s} = \frac{(Y_0 - Y_1)}{\left[\frac{Y_1}{m}\right]^{1/n}} \rightarrow (6)$$

sub x_2 in equn (2)

$$L_s (Y_1 - Y_2) = S_{s_2} \left(X_0 - \left(\frac{Y_2}{m}\right)^{1/n}\right)$$

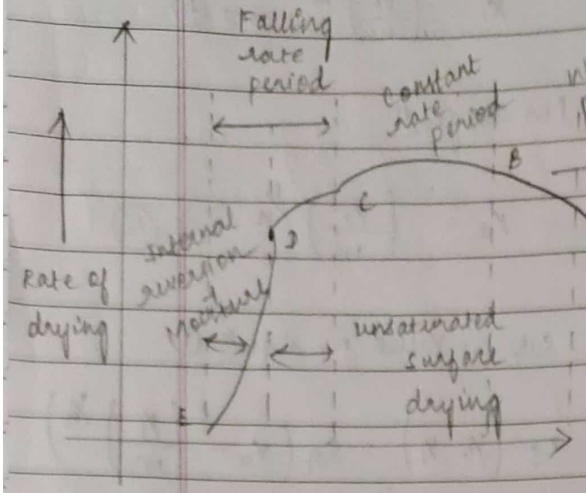
$$\frac{S_{s_2}}{L_s} = \frac{(Y_1 - Y_2)}{\left(\frac{Y_2}{m}\right)^{1/n}} \rightarrow (7)$$

for total adsorbent used

$$\left(\frac{S_{s_1} + S_{s_2}}{L_s}\right) = m^{1/n} \left(\frac{Y_0 - Y_1}{Y_1^{1/n}} + \frac{Y_1 - Y_2}{Y_2^{1/n}}\right) \rightarrow (8)$$

Drying:

Removal of Moisture from the substance by using heat application



Critical Moisture content

Moisture content of the Material at which the constant rate period ends & Boiling rate periods Start. It is the func of the constant drying rate material propotion

Moisture evoprated per unit time.

Where the rate of drying various looking time

Time of drying under constant rate of drying condition.

$t = t_c + t_f \longrightarrow ①$

t — Total time of Drying

t_c — constant rate time

t_f — Falling time.

a) constant rate period:

$x_1 \rightarrow$ Initial Moisture content.

$x_2 \rightarrow$ Final

$x_c \rightarrow$ Critical.

$$\text{Rate of drying} = \frac{-W'}{A}\frac{dx}{dt} \hookrightarrow ②$$

$W' \rightarrow$ Wt. of the solid

$A \rightarrow$ Drying area

$\frac{dx}{dt} \rightarrow$ change in Moisture w.r.t time

For constant rate (R_c)

$$R_c = -\left(\frac{W'}{A}\right)\left(\frac{dx}{dt}\right) \rightarrow ③$$

rearrange equn ②

$$dt = \frac{-W'}{A R_c} \cdot dx \longrightarrow ④$$

Integrate equn ④

$t \rightarrow 0 - t$

$x \rightarrow x_1 \rightarrow x_2$

$$\int_0^t dt = \left(\frac{-W'}{A R_c}\right) \int_{x_1}^{x_2} dx$$

$$t = \left(\frac{-W'}{R_c A}\right)(x_2 - x_1)$$

$$t = \frac{W'}{R_c A}\left[x_1 - x_2\right] \longrightarrow (5)$$

For moisture x_c;

$$t_c = \frac{W'}{A R_c}(x_1 - x_c) \longrightarrow (6)$$

b) Falling rate period:

Rate of drying $\propto$ moisture content

$$-\frac{W'}{A}\frac{dx}{dt} \propto (x - x^*) \longrightarrow (7)$$

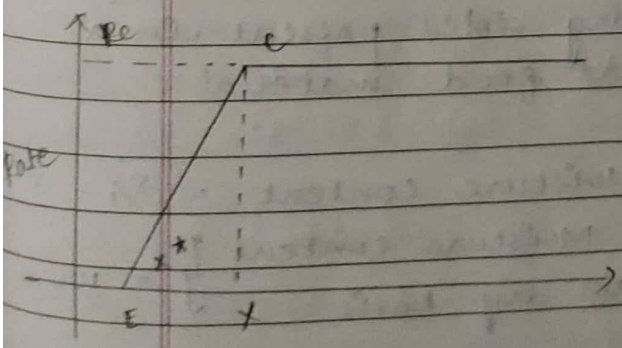
$$\frac{-W'}{A}\frac{dx}{dt} = m(x - x^*) \longrightarrow (8)$$

$x^* \longrightarrow$ equivalent Moisture content.

$x \longrightarrow$ central Moisture content

$x_1 \longrightarrow$ Initial

$x_2 \longrightarrow$ Final



C_E - line $\longrightarrow$ Falling rate period. m = slope.

$$m = \frac{R_c}{(x_c - x^*)} = \frac{dy}{dx} \longrightarrow (9)$$

rearrange equn (8)

$$\frac{-w'dx}{A.dt} = \frac{R_c}{(x_c - x^*)}(x - x^*)$$

$$\frac{-dx}{(x - x^*)} = \frac{R_c}{(x_c - x^*)w}dt \longrightarrow (10)$$

Integrate equn (10) with

$t \longrightarrow 0 \longrightarrow t$

$x \longrightarrow x_1 \longrightarrow x_2$

$$-\int_{x_1}^{x_2}\frac{dx}{(x - x^*)} = \frac{R_c A}{[x_c - x^*]w'}\cdot\int_0^t dt$$

$$t = \frac{w'(x_c - x^*)}{R_c A}\ln\left(\frac{x_1 - x^*}{x_2 - x^*}\right)$$

$$t_f = \frac{w'(x_c - x^*)}{R_c A}\ln\left(\frac{x_1 - x^*}{x_2 - x^*}\right) \longrightarrow (10)$$

Total time of drying $\Big\} t = t_c + t_f$

$$t = \left[\frac{W'}{RA}(x_1 - x_2)\right] + \left[\frac{W'(x_c - x^*)}{R_c A}\ln\left(\frac{x_1 - x^*}{x_2 - x^*}\right)\right] \longrightarrow (11)$$

1) A 50 kg batch of granular solids containing 25 % Moisture is to be dryed in a tray dryer to 12 % Moisture by passing a steam @ 363 K (90°C) tangentially across the surface @ velocity of 1.8 m/s if const. rate of drying under these condition is 0.0008 kg/hr moisture & critical Moisture content is 10 %. calculate the drying time. The surface area available for drying is $1 m^2$.

Given data:

on wet basis

$x_1 = 25\% = 0.25$

$x_2 = 12\% = 0.12$

Critical Moist. content = 10% = 0.1 (x_c)

$R_c = 0.0008$ kg moist / $m^2.s$

$A = 1 m^2$

Soln: on dry basis

$w = 50(0.75)$

$w = 37.5$

$x_c < x_2$

Time for constant rate period

$$t = \frac{-W}{A}\left(\frac{x_1 - x_2}{R_c}\right)$$

$$x_1 = \frac{x_1}{1-x_1} = \frac{0.25}{1-0.25} = 0.33$$

$$x_2 = \frac{0.12}{1-0.12} = 0.136$$

$$R_c = 0.0008 \frac{kg}{m^2 s}\left(\frac{3600\ s}{1\ h}\right)$$

$$R_c = (0.0008 \times 3600)\ kg/m^2h$$

$$R_c = 2.88\ kg/m^2h.$$

$$t = \frac{(37.5)}{1}\left(\frac{0.33 - 0.136}{2.88}\right)$$

$$t = 2.526\ hr$$

2 Marks:

1) Moisture content: Wet basis

It is expressed as the ratio of the wt of the moisture to the wt of the wet feed material moisture content on wet basis = $\frac{x}{1+x}$

Wt % moisture content $= 100 \cdot \frac{x}{1+x}$

2) Moisture content on dry basis:

It is expressed as the ratio of the wt of the moisture to the wt of the dry solid present in a wet feed material.

Moisture content $= x/1 = x$

% Moisture content on dry basis $= 100x$

3) Equilibrium Moisture content:

The Moisture content of a solid material that is the thermodynamic equilibrium with its vapour in the gas phase under the specified humidity & temp of the gas or air is termed as equi. Moisture content.

4) Bound Moisture Content:

It is that moisture in a solid material which exerts a vapour pressure < than that of the pure liq. at the same temp.

5) Unbound Moisture Content:

It is the moisture held by a solid material in excess of the equilibrium humidity. It is primarily held in the voids of solid.

Equi vapour pressure in a solid material in equal to the pure liq at the give temp.

6) Critical Moisture content:

The Moisture content of a Material at which the constant rate period ends & the following rate period starts is called critical moisture content. It is the func. of the constant drying rate material proportion & partial size.

7) constant rate period:

It is that part of the drying process during which the rate of drying pressed as the moisture evaporated per unit time per unit area of drying surface remains constant.

8) Falling rate period:

It is that part of the drying process during which the rate of drying varies with time & the instantaneous drying rate continuously decreases.

9) Relative humidity:

Ratio of partial pressure of water vapour in the air water vapour mixture

$$\% RH = \frac{P_A}{P_A^\circ} \times 100$$

$P_A \longrightarrow$ partial pressure of H_2O vapour in mixture

$P_A^\circ \longrightarrow$ Vapour pressure of pure H_2O.

10. Humidity:

$$H = \frac{\text{kg of } H_2O \text{ vapour}}{\text{kg of dry air}} = 0.62\ Hm$$

Hm = Molal humidity.

11. Dry bulb temp:

The temp of the vapour gas mix recorded by a thermometer whose bulb is kept dry.

12. Wet bulb temp:

The temp recorded by a thermometer whose bulb is kept wet by wrapping a wet cloth in the open air is called as the wet bulb temp.

13. Dew pt:

The temp at which the condensation will first occur is known as dew point.

5/11/22

2) The eqw. Adsorption of Acetone vapour on activated carbon is given by

$\frac{g \text{ adsorbed}}{g \text{ carbon}}$	0	0.1	0.2	0.3	0.35
Partial pressure of acetone (mmHg)	0	2.0	12.0	42.0	92.0

1 l flask contains a acetone vapour at 1 atm. pressure and 30°C with

Soln:

Mol.Wt of acetone = 58

Air ⇒ 79% N_2 21% O_2

Mol. Wt. of air = [0.79 (28) + (0.21)

= 28.84

Total pressure = 1 atm = 760 mm

P.P of acetone vapour = 100 mm

P.P of air = Total pres − P.P of acetone

= 760 − 100 = 660 mmHg

$$Y_0 = \frac{[\text{P.P of acetone}]}{\text{P.P of air}} = \frac{100 \times 58}{(660 \times 28.84}$$

$Y_0 = 0.3047$

By heating the P.P of the air, we can compute the P.P of acetone in the flask.

$$Y_1 = \frac{(0) \times (58)}{((660) \times 28.84)} = 0$$

$$Y_2 = \frac{(2 \times 58)}{(660 \times 28.84)} = 0.006$$

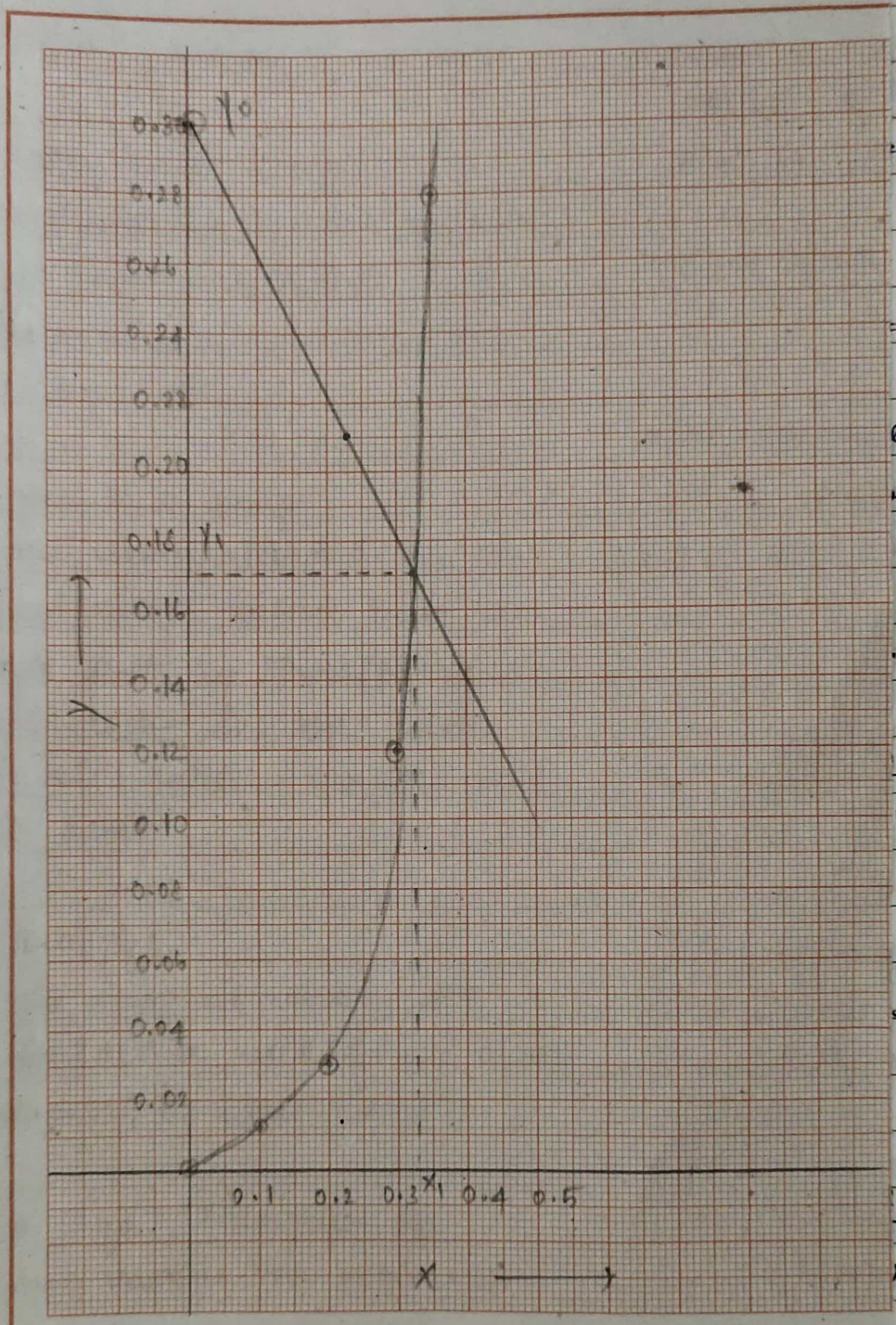
$$Y_3 = \frac{12 \times 58}{(660 \times 28.84)} = 0.036$$

10. Humidity:

$H = \frac{\text{kg of } H_2O \text{ vapour}}{\text{kg of dry air}} = 0.62$

vapour at 1 atm. pressure and 30°C with

(2)



acetone = 58

:9% N_2 21% O_2

$= [0.79(28) + (0.21)(3$

$= 28.84$

e = 1 atm = 760 mmHg

ne vapour = 100 mmHg

Total pres - p.p of acetone

$100 = 660$ mmHg

acetone] $= \frac{100 \times 58}{(660 \times 28.84}$

of air

.3047

the p.p of the

can compute

acetone in the

$\frac{(58)}{} = 0$

$\times 28.84)$

$\frac{(2 \times 58)}{} = 0.006$

(660×28.84)

g carbon					
Partial pressure of acetone (mmHg)	0	2.0	12.0	42.0	92.0

$Y_3 = \frac{12 \times 58}{(660 \times 28.84)} = 0.036$

1 l flask contains a acetone

$$y_4 = \frac{42 \times 58}{660 \times 28.84} = 0.127$$

$$y_5 = \frac{92 \times 58}{660 \times 28.84} = 0.2803$$

x	0.1	0.2	0.3	0.35
y	0	0.006	0.036	0.127	0.2803

Oil passed through (x_0, y_0) & (x_1, y_1)

$$\text{Slope} = \frac{-S_s}{L_s}, \quad L_s = G_s = \text{air in the flask}$$

$$n = \frac{PV}{RT} = 30°C = 303K$$

$V = 1L = 1000\ cc$

$P = 660\ mmHg \qquad R = 82.06 \frac{m^3 atm}{g\,mol\,k}$

$n = 0.034\ g\,mol.$

Mass of air $= (28.84)(0.034)$

$G_s = 0.9805\ g \qquad S_s = 2g$

$$\text{Slope} = \frac{-S_s}{G_s}$$

$$\text{Slope} = \frac{-2}{0.9805} = -2.039$$

$$\theta = \tan^{-1}(-2.039)$$

$$\theta = 63.87$$

From graph,

Amount of Acetone adsorbed (x_1) } $= 0.32$

Amount of Acetone in vapour (y_1) } $= 0.16$

Experiment of decolourisation on oil heated use following equ. relⁿ $y = 0.5x^{0.5}$ where y is the g of colour removed per g of adsorbent where x is the coloured in the oil 100 kg oil containing 1 part of colour to the 3 parts of oil (1:3) is agitated with 25 kg of adsorbent calculate % of coloured removed, if i) All 25 kg adsorbent is used in 1 step. ii) 12.5 kg adsorben is used initially followed by another 12.5 kg of adsorbent.

$$Y = \frac{g \cdot \text{colour}}{g \text{ colour free oil}}.$$

$$x = 1000y$$

$$x = \frac{g\ \text{colour}}{g\ \text{of adsorbent}}$$

$$X = y$$

Rearrange eq. relationship

$$Y_1 = 0.5x^{0.5}$$

$$x = 0.5(1000y)^{0.5}$$

$$y = 0.004x^2$$

A - oil, B = adsorbent, C = colour

100 kg of oil.

a) L_s = Mass of colour free oil

$= 75$ kg

$S_s = 25$ kg.

$Y_0 = \frac{\text{mass of c}}{\text{mass of A}} = \frac{25}{75} = 0.333$

$L_s(Y_0 - Y_1) = S_s(X_1 - X_s)$

X & Y_1 are equ. conc^n. $X_0 = 0$

$Y_1 = 0.004 X_1^2$

$75\left[0.333 - (0.004 X_1^2)\right] = 25(X_1)$

$24.47 - 0.3 X_1^2 = 25 X_1$

$X_1 = 0.988$

% colour removed $= \frac{25 \times X_1}{25} \times 100$

$= 98.8\%$

b) 2 stage adsorption

For each stage 12.5kg adsorbent used

Stage I: $S_s = 12.5$, $X_0 = 0$

MB:

$L_s(Y_0 - Y_1) = S_s(X_1 - X_0)$

$75(0.333 - (0.004 X_1^2) = 12.5 X_1$

$X_1 = 1.974$

$Y_1 = 0.004 X_1^2$

$Y_1 = 0.004 (1.974)^2$

$Y_1 = 0.0155$

Stage II

M.B equn

$\Rightarrow L_s(Y_1 - Y_2) = S_s(X_2 - X_0)$

$Y_2 = 0.004 X_2^2$

$75\left[(0.015 - (0.004 X_2^2)\right] = 12.5 X_2$

$X_2 = 0.066$

$Y_2 = 1.742 \times 10^{-5}$

% colour removed $= \left(\frac{Y_0 - Y_2}{Y_0}\right) \times 100$

$= \left(\frac{0.333 - 1.742 \times 10^{-5}}{0.333}\right) \times 100$

$= 423.12\%$